# Semiparametric Dynamic Logit Model with Endogenous Networks

Brice Romuald Gueyap Kounga

Department of Economics, Western University

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# Many choices persist over time and are made among friends

- Teenagers who smoke this year often smoke next year, and their friends often smoke too (Fletcher, 2010; Nakajima, 2007).
- Farmers who adopt a new technology this season often keep it, and farmers who exchange information often make the same decision (Conley and Udry, 2010).

## Two mechanisms behind the same pattern

- The current choice may depend on the past choice: state dependence.
- Unobserved characteristics (e.g., a taste for risk, impulsivity, patience, or ability) may drive the choice in every period, and the same characteristics shape their choice of friends.

To study such choices, the econometrician should use a model in which the network is endogenous: links and outcomes depend on the same unobserved characteristics.

# The question: habit or unobserved characteristics?

Why does smoking persist from one year to the next?

## State dependence

Smoking last year raises the chance of smoking this year.

A one-year intervention that stops a student from smoking also lowers her probability of smoking in the following years (Heckman, 1981a).

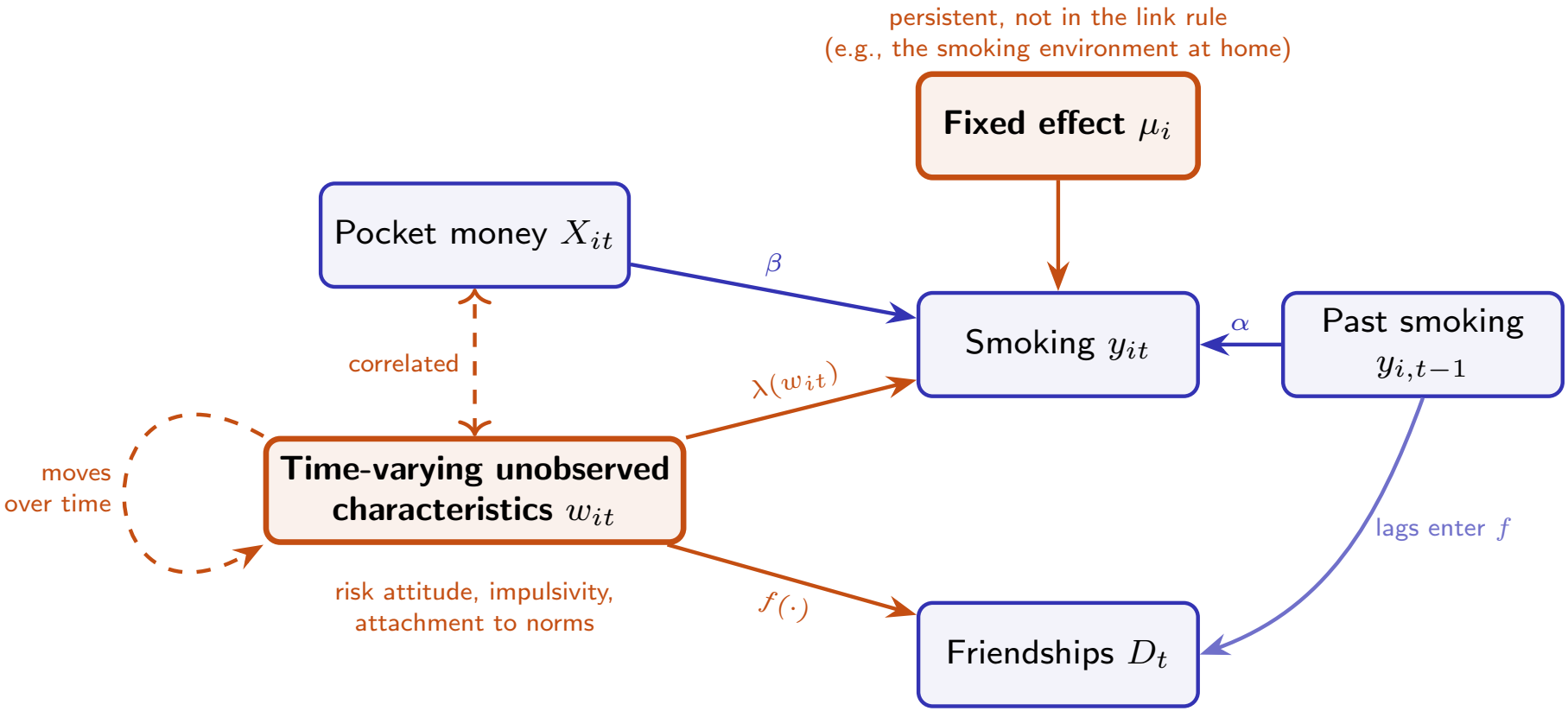
## Unobserved characteristics

A taste for risk or impulsivity makes the same student smoke in every wave.

The same intervention has no lasting effect: once it ends, the characteristics are still there.

- The two readings call for different policies, so the parameter that separates them, $\alpha$, matters.
- The same characteristics also shape students' choice of friends, and some of them change during adolescence, which is what makes the problem hard.

# The confounding problem




- The same unobserved characteristics drive the choice, the friendships, and the persistence, and the researcher has no data on them.
- Estimates of $\beta$ and $\alpha$ that ignore this stay biased however large the sample.

# Existing methods do not resolve this problem, for different reasons

- **Pooled dynamic logit.** The effect of the unobserved characteristics is loaded onto pocket money and past smoking.
- **Controlling for friends' characteristics.** Peer averages recover only the part of the unobserved characteristics that is reflected in friends' *observables*.
- **Fixed effects with conditional likelihood** (Chamberlain, 1980; Honoré and Kyriazidou, 2000). Remove a characteristic that never changes; the part that moves during adolescence stays in the comparison.
- **Control functions and parametric network models** (Goldsmith-Pinkham and Imbens, 2013; Hsieh and Lee, 2016; Johnsson and Moon, 2021). Each rests on restrictions on how links form that differ from the ones used here.

This paper restricts the arguments of the link rule but leaves its functional form unknown.

# This paper

## Idea

Use the observed friendship network to compare two students of the same network type: the same probability of befriending each other student, given past smoking. Within each student, a switch in smoking removes the fixed effect and the level of the unobserved influence. Across the two students, the comparison removes its change, provided the change is common within a network type. No parametric link model is assumed.

1. **Identification** of the slope $\beta$ and the state dependence $\alpha$ in a semiparametric dynamic logit with nonparametric link formation, in which a time-varying unobserved characteristic affects outcomes and links, alongside a fixed effect.
2. **Estimation and inference.** A pairwise conditional likelihood estimator that matches agents on an estimated codegree distance: consistent, including with continuous types, and root-$n$ normal when the characteristic takes finitely many values and matches are exact, also with a data-driven bandwidth when $T = 3$. A second estimator conditions on recovered network classes.
3. **Evidence.** Simulations in designs that satisfy the assumptions and designs that violate them, and a feasibility exercise on adolescent smoking in Glasgow that documents what the method needs from data.

# Preview of results

## Simulations ($n$ from 500 to 3000, four latent types that move between waves)

- The bias of the pairwise estimator falls with the sample size and its intervals cover the truth in 94 to 97 percent of the replications that return an estimate; it agrees with an oracle that knows the true matches.
- The within-agent estimator of Honoré and Kyriazidou (HK) keeps a bias of 0.23 to 0.35 for $\beta$, and its coverage falls to 0.75 by $n = 3000$; pooled and peer-control logits are biased throughout.
- The class-conditional estimator reduces the RMSE of the pairwise estimator by 6 to 20 percent in the two main designs.

## Application (96 students, three waves, 22 of whom switch)

- Descriptive fits give a positive point estimate of state dependence in each of the five unlocalized fits, falling from 2.22 in a pooled logit to 0.89 under class conditioning; the estimates are imprecise and the ordering is sensitive to the penalization.
- When the network is used to select matched pairs, every localized likelihood is completely separated, so the exercise documents what the method needs from data.

# Related literature

- **Dynamic and panel logit.** Chamberlain (1980, 2010); Honoré and Kyriazidou (2000); Honoré and Tamer (2006); Honoré and Weidner (2025); Honoré, Muris, and Weidner (2025); Dano, Honoré, and Weidner (2025); Shin (2025).
  Unobserved heterogeneity is mostly a time-invariant fixed effect; time variation only through structures in observed indices, such as trends.

- **Peer effects and endogenous networks.** Manski (1993); Goldsmith-Pinkham and Imbens (2013); Hsieh and Lee (2016); Arduini, Patacchini, and Rainone (2015); Badev (2021); Johnsson and Moon (2021); Kashaev and Lazzati (2025); Graham (2017); Zeleneev (2020); Gao and Niu (2026).
  A parametric link model, equilibrium restrictions, or control functions; or the formation model itself.

- **Matching on network types.** Auerbach (2021, 2022) for partially linear regression; Gueyap Kounga (2023) for a static semiparametric logit; Starck (2025) and Xu (2025) for network-based controls.
  A single cross-section: equal levels of the unobserved term within a network type.

This paper: a dynamic logit with a fixed effect and a time-varying unobserved characteristic that also drives a changing network, with equal *changes* within a network type rather than equal levels, and no parametric model of link formation.

# Model

Panel of $n$ agents observed at $t = 0, 1, \ldots, T$ with $T \geq 3$; the initial outcome $y_{i0}$ is observed.

$$\text{Outcome:} \quad y_{it} = \mathbb{1}\{X_{it}'\beta + \alpha\, y_{i,t-1} + \lambda(w_{it}) + \mu_i - \varepsilon_{it} \geq 0\}, \qquad t = 1, \ldots, T$$

$$\text{Network:} \quad D_{ijt} = \mathbb{1}\{f(w_{it}, w_{jt};\, y_{i,t-1}, y_{j,t-1}) \geq \eta_{ijt}\}\, \mathbb{1}(i \neq j)$$

| | |
|---|---|
| $X_{it}$ | time-varying covariates, such as pocket money (no intercept, which $\lambda$ and $\mu_i$ absorb) |
| $y_{i,t-1}$ | past outcome; $\alpha$ measures state dependence |
| $w_{it} \in [0, 1]$ | index of unobserved characteristics, varying across agents and over time |
| $\lambda(\cdot)$ | the social influence function (Auerbach, 2022): unknown and unrestricted in shape |
| $\mu_i$ | fixed effect: persistent characteristics that affect the outcome but are not arguments of $f$ |
| $f(\cdot)$ | unknown, jointly symmetric link function; links may depend on past outcomes |
| $\varepsilon_{it},\ \eta_{ijt}$ | logistic outcome shocks and uniform link shocks |

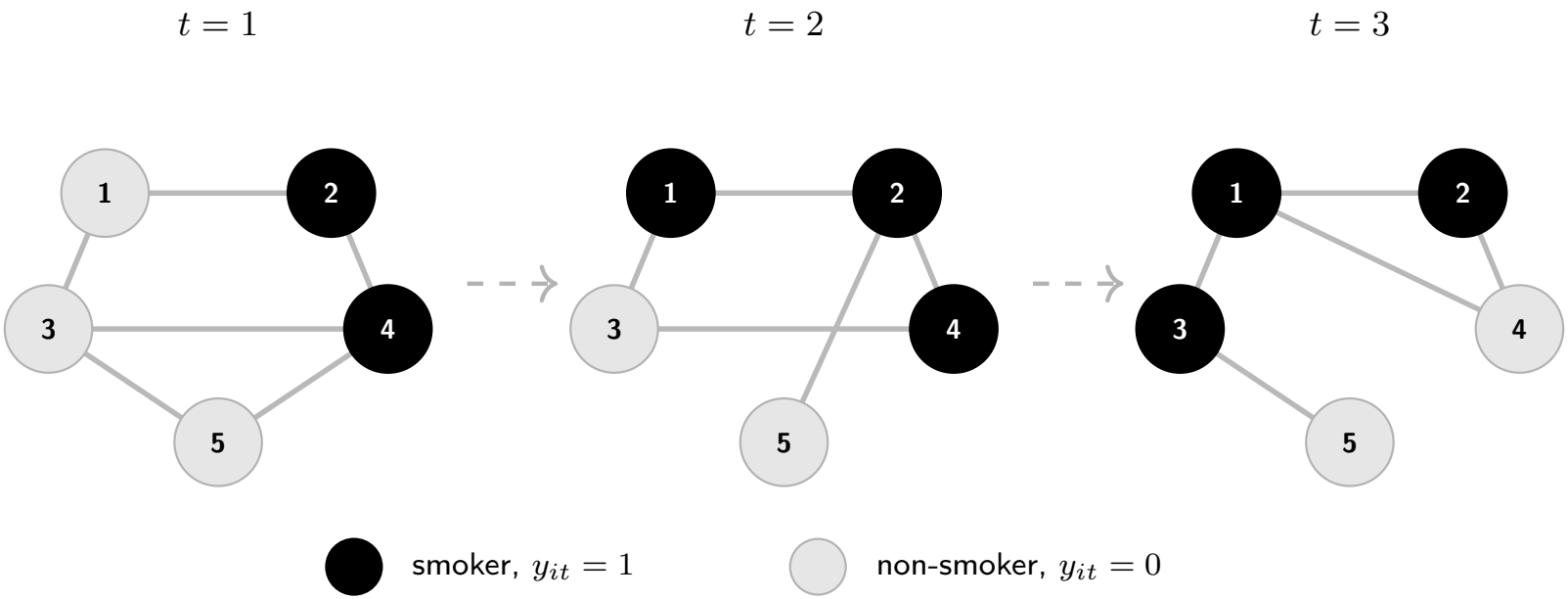
Parameters of interest: $(\beta, \alpha)$. Nuisance components: $\lambda$, $f$, and $\mu_i$, which the method removes rather than estimates.

▸ Assumption 1

## What the social influence function is, and what it is not

- $\lambda(w_{it})$ is the agent's own pull toward the outcome exerted by the characteristics that also shape her ties. It is not an effect of friends' smoking: the equation contains no friend's outcome and no term in the realized network, and the paper identifies no peer effect.
- The network enters only as information about $w_{it}$, because the same $w_{it}$ governs her choice of friends.
- **Two components of unobserved heterogeneity, removed by two arguments.**
  - $\mu_i$ and the *level* of $\lambda(w_{it})$ cancel within each agent over time, as in Chamberlain (1980) and Honoré and Kyriazidou (2000);
  - the *change* of $\lambda(w_{it})$ over time, which a within-agent comparison cannot remove, is removed by comparing agents of the same network type.
- Agents of the same network type need not share $\mu_i$, and $\mu_i$ may depend on $(X_i, w_i)$ and $y_{i0}$ in an arbitrary way. The level of $\lambda$ is never recovered: $\mu_i$ can absorb any constant.
- **Special cases.** $\lambda \equiv 0$: the dynamic logit with fixed effects of Honoré and Kyriazidou (2000). $\mu_i \equiv 0$, $\alpha = 0$, $w_{it} = w_i$, and links that do not depend on past outcomes: the static semiparametric logit of Gueyap Kounga (2023).

# Outcomes and friendships evolve together



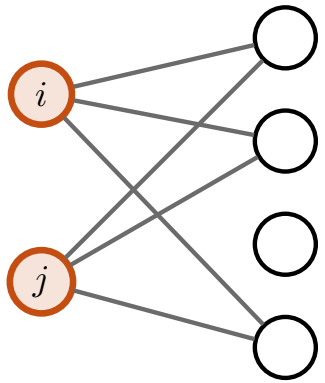

- Agent 1 starts smoking between $t = 1$ and $t = 2$ and keeps smoking, which is the pattern state dependence produces.
- Agent 3 starts at $t = 3$ as her unobserved characteristics shift toward risk-taking, and agent 4 quits at $t = 3$ while forming a new link with agent 1.
- The researcher observes outcomes and links at every wave, but never the $w_{it}$ or $\mu_i$ that drive them.
- The picture illustrates the general model, before any restriction on how $w_{it}$ moves.

# Network type: how an agent forms links

For each pair of lagged outcomes $(y, y')$, the conditional individual link function of agent $i$ is

$$f_{w_{it};\, y,y'}(s) = f(w_{it}, s;\, y, y'), \qquad s \in [0,1],$$

the probability that an agent with index $w_{it}$ and lagged outcome $y$ links with an agent of characteristic $s$ and lagged outcome $y'$.




$i$ and $j$ link with every kind of partner with the same probability

A student's network type is the list of her probabilities of befriending each other student, given her past smoking.

For two agents who share the lagged outcome $y$, the network-type distance is

$$\rho_{ijt} = \Big[ \sum_{y' \in \{0,1\}} \int_0^1 \big( f_{w_{it};\, y,y'}(s) - f_{w_{jt};\, y,y'}(s) \big)^2 ds \Big]^{1/2}.$$

- $\rho_{ijt} = 0$ means that $i$ and $j$ have the same network type.
- "Type" describes linking behavior, not private information as in game theory; two agents of the same type may have different values of $w_{it}$.

# The identifying assumption: equal changes, not equal levels

## Assumption 2 (latent index dynamics)

(i) **Movement common to network types.** For each $t \leq T-1$, $\lambda(w_{i,t+1}) - \lambda(w_{it}) = d_t(w_{it})$, with $d_t$ uniformly continuous in the network-type distance, hence constant on network-type classes; its magnitude is unrestricted.

(ii) **Stability after the transition.** For each transition period $s$ used, $d_{s+1} \equiv 0$.

- Agents of the same network type may differ in $\lambda(w_{it})$ at every date and in $\mu_i$; what they share is the movement of $\lambda$ between consecutive dates.
- Weaker than equal values of $\lambda$ within a type (Auerbach, 2022; Gueyap Kounga, 2023), which allows a fixed effect that matching does not equalize.
- Plausible when the same characteristics drive the outcome and the ties, so that a change shows up in both; it fails when the movement differs within a type, as with idiosyncratic innovations. A movement common to every agent is allowed. Since $\lambda$ is never observed, the assumption cannot be tested from the data alone.
- Maintained throughout (Assumption 1): strict exogeneity. Neither the covariates nor $w_{it}$ respond to past outcome shocks, which excludes, for example, drinking that responds to past smoking.

# When does the assumption hold?

| Process for the characteristics | Status under Assumption 2(i) |
|---|---|
| Time-invariant, $w_{it} = w_i$ | $d_t \equiv 0$: holds trivially |
| Common map $w_{i,t+1} = g_t(w_{it})$ | holds if $\lambda \circ g_t - \lambda$ is constant on each class and uniformly continuous; a class-preserving map does not guarantee it ▸ Example |
| Simulation designs: types move between network blocks by a common permutation | holds; every class is a single type |
| Idiosyncratic movement of fixed size (e.g., a stationary autoregression) | fails: no conditioning removes it |
| Idiosyncratic movement that shrinks with $n$ | covered by the asymptotic theory (Assumption 3) |

Part (ii), no movement after the transition, must hold in every case.

## The observation window in the smoking example

Four dates $t = 0, 1, 2, 3$ around a scheduled school transition between $t = 1$ and $t = 2$. A special case:

$$\lambda(w_{i1}) = \ell_{c(i)}, \qquad \lambda(w_{i2}) = \lambda(w_{i3}) = \ell_{c(i)} + \Delta_{c(i)},$$

with $c(i)$ the student's network class before the transition. Students who formed friendships in the same way share the change $\Delta_c$, while their fixed effects can differ; the adjustment is complete by $t = 2$; and the dates are comparably spaced. ▸ Details

## Step 1: compare the two orderings of one agent's outcomes

For an interior period $s$, take two outcome paths that differ only at $s$ and $s+1$:

$$A_s = (\ldots, y_{i,s-1}, \underbrace{0,1}_{s,\,s+1}, y_{i,s+2}, \ldots), \qquad B_s = (\ldots, y_{i,s-1}, \underbrace{1,0}_{s,\,s+1}, y_{i,s+2}, \ldots).$$

### Lemma 1 (within-agent likelihood ratio)

Conditional on $\chi_i = (y_{i0}, X_i, w_i, \mu_i)$ with $X_{i,s+2} = X_{i,s+1}$ and $\lambda(w_{i,s+2}) = \lambda(w_{i,s+1})$,

$$\frac{\mathbb{P}(y_i \in B_s \mid \chi_i)}{\mathbb{P}(y_i \in A_s \mid \chi_i)} = \exp\Big[(X_{is} - X_{i,s+1})'\beta + \alpha(y_{i,s-1} - y_{i,s+2}) + \underbrace{\lambda(w_{is}) - \lambda(w_{i,s+1})}_{\text{still here}}\Big].$$

The ratio does not depend on $\mu_i$.

- The fixed effect and the level of $\lambda$ drop out, as in Chamberlain (1980) and Honoré and Kyriazidou (2000), but the change of $\lambda$ between $s$ and $s+1$ remains.
- The condition at $s+2$ is needed because swapping the outcomes at $(s, s+1)$ changes the lag that enters at $s+2$.

▸ Why four dates

## The case $T = 3$ in one display

With $c_{it} = X_{it}'\beta + \lambda(w_{it}) + \mu_i$ and $\Gamma(c) = \log(1 + e^{c+\alpha}) - \log(1 + e^{c})$, and without any stability condition,

$$\log \frac{\mathbb{P}(y_i \in B_1 \mid \chi_i)}{\mathbb{P}(y_i \in A_1 \mid \chi_i)} = (X_{i1} - X_{i2})'\beta + \alpha(y_{i0} - y_{i3}) + \underbrace{\left[\lambda(w_{i1}) - \lambda(w_{i2})\right]}_{\text{movement between the transition waves}} + \underbrace{\left[\Gamma(c_{i3}) - \Gamma(c_{i2})\right]}_{\text{period-3 term}}.$$

- **The fixed effect** cancels in $c_{i1} - c_{i2}$, which is the covariate term plus the first bracket; it enters the period-3 term only through the level $c_{i2}$, unless $c_{i3} = c_{i2}$.
- **Period-3 term.** Since $|\Gamma'(c)| < 1$, it is bounded by $\|\beta\| \, \|X_{i3} - X_{i2}\| + |\lambda(w_{i3}) - \lambda(w_{i2})|$ and vanishes when covariates and social influence are stable between waves 2 and 3: the covariate kernel $K_2$ and Assumption 2(ii) target it.
- **Movement between the transition waves.** Within-agent conditioning can remove it only by assuming it away, as the fixed-effect specification of HK does. Cross-agent matching removes it without that assumption.

## Step 2: compare two agents of the same network type

Take agents $i$ and $j$ with the same lagged outcome at $s-1$, who switch in opposite directions between $s$ and $s+1$ (one starts, the other quits), who have the same network type at $s$, and whose covariates are stable between $s+1$ and $s+2$: the observable event $\mathcal{O}_{ijs}$, with the discordance event $\mathcal{D}_{ijs}$.

### Theorem 1 (identification of $\beta$ and $\alpha$)

Under Assumptions 1 and 2, if $\mathbb{P}(\mathcal{O}_{ijs} \cap \mathcal{D}_{ijs}) > 0$ and the same type implies the same change,

$$\mathbb{P}\big(y_i \in A_s,\, y_j \in B_s \,\big|\, \mathcal{F}_{ijs},\, \mathcal{O}_{ijs},\, \mathcal{D}_{ijs}\big) = F\big[(\Delta_s X_j - \Delta_s X_i)'\beta + \alpha(y_{j,s-1} - y_{j,s+2} - y_{i,s-1} + y_{i,s+2})\big],$$

with $\Delta_s X_i = X_{is} - X_{i,s+1}$, $F$ the logistic distribution function, and $\mathcal{F}_{ijs}$ the covariates and the outcomes outside the window. With a rank condition on the regressors over the selected pairs, $(\beta, \alpha)$ is identified; with continuous types, in the limit of shrinking neighborhoods $\mathcal{O}_{ijs}(h)$.

- Same type at $s$ gives $\lambda(w_{is}) - \lambda(w_{i,s+1}) = \lambda(w_{js}) - \lambda(w_{j,s+1})$ by Assumption 2(i), so every term involving $\lambda$ and $\mu$ cancels, whatever the size of the change.
- The fixed effects never enter, and agents of the same type need not share them.

▸ Conditioning event and proof

# Two comparisons, two jobs

Within one agent, over time → removes the fixed effect and the level of $\lambda$, which are constant for the agent

Across two agents with the same network type → removes the common change $d_s$, whatever its size

- If $d_s \equiv 0$, within-agent conditioning alone is valid: $\lambda(w_i) + \mu_i$ is then a fixed effect and the network adds nothing.
- The identifying power of the network lies in the common component: it cancels across matched agents but enters the within-agent ratio as an omitted term $d_s(w_{is})$.
- **Corollary 1.** Given $(\beta, \alpha)$, the common movement $d_s$ of each network class is identified from the odds of switching down rather than up among its switchers; the level of $\lambda$ is not.

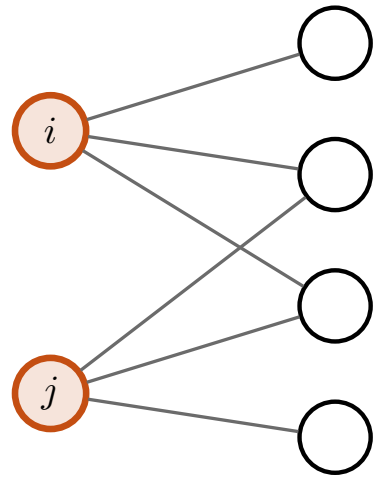
▸ What is identified about $\lambda$

# From network type to something computable: codegree distance

The network type is a population object, but the data are the adjacency matrices $D_t$.

$$\hat{\delta}_{ijt} = \left[\frac{1}{n}\sum_{k=1}^{n}\left(\frac{1}{n}\sum_{l=1}^{n} D_{klt}\left(D_{ilt} - D_{jlt}\right)\right)^2\right]^{1/2}$$

The codegree of two students is their number of common friends. For each third student $k$, compare the codegree of $(i,k)$ with the codegree of $(j,k)$; square the difference and average over $k$.




$i$ and $j$ share two of their three friends

- Computed from the observed networks alone, without specifying the link function (Auerbach, 2022).
- **Proposition 1.** Uniformly over pairs that share the lag, $\max_{i,j} |\hat{\delta}^2_{ijt} - \delta^2_{ijt}| = O_p\left((\log n/n)^{1/2}\right)$; under continuity and full support, $\delta_{ijt} = 0$ if and only if the two agents have the same network type; when the characteristic takes finitely many values, every positive squared distance is at least a gap $\bar{\epsilon}$.

▸ Population distance

## Lag stratification

- Links depend on lagged outcomes, so the neighborhoods of $i$ and $j$ are comparable only when $y_{i,t-1} = y_{j,t-1}$; otherwise the codegree difference mixes the type difference with the effect of the lag.
- The aggregated lag-stratified distance averages over the periods where the two agents share the lag:

$$\hat{\bar{\delta}}^2_{ij} = \frac{\sum_{t=1}^{T} \mathbb{1}\{y_{i,t-1} = y_{j,t-1}\}\, \hat{\delta}^2_{ijt}}{\max\left(1, \sum_{t=1}^{T} \mathbb{1}\{y_{i,t-1} = y_{j,t-1}\}\right)}, \qquad \hat{\bar{\delta}}^2_{ij} = +\infty \text{ if they never share it.}$$

- For a pair that switches in opposite directions, periods $s+1$ and $s+2$ never qualify because the lagged outcomes differ there. Equivalence is certified at period $s$; the equal change between $s$ and $s+1$ comes from Assumption 2(i), not from the data.
- With $T = 3$ and $s = 1$, the only informative period for a discordant pair is $t = 1$, and the aggregated distance is $\hat{\delta}^2_{ij1}$.

### In the simulations

When links depend on lagged outcomes (the lag-dependent design), the stratified distance still separates the four types exactly, and the results are those of the baseline design.

## The pairwise estimator

Pairs that share the lagged outcome and switch in opposite directions between $s$ and $s+1$:

$$\mathcal{P}_s = \big\{(i,j):\ i<j,\ y_{i,s-1}=y_{j,s-1},\ y_{is}+y_{i,s+1}=1,\ y_{js}+y_{j,s+1}=1,\ y_{is}\neq y_{js}\big\}.$$

## Kernel-weighted conditional maximum likelihood

$$(\hat\beta,\hat\alpha)=\arg\max_{(b,a)}\sum_{s\in\mathcal{S}}\sum_{(i,j)\in\mathcal{P}_s}\underbrace{K_1\Big(\frac{\hat{\bar\delta}_{ij}^2}{h_1}\Big)}_{\text{networks alike}}\underbrace{K_2\Big(\frac{\|X_{i,s+2}-X_{i,s+1}\|^2}{h_2}\Big)K_2\Big(\frac{\|X_{j,s+2}-X_{j,s+1}\|^2}{h_2}\Big)}_{\text{covariates stable after the switch}}m_{ijs}(b,a)$$

- $m_{ijs}$ is the logistic log-likelihood of which agent started and which quit, with index $(\Delta_s X_i-\Delta_s X_j)'b+a(y_{i,s-1}-y_{i,s+2}-y_{j,s-1}+y_{j,s+2})$, so the objective is concave.
- $K_2$ comes from the dynamics, as in HK, and serves only the covariate condition of Lemma 1. The stability of $\lambda$ after the transition comes from Assumption 2(ii).
- With $T=3$, a bandwidth computed from the first-wave network keeps the distribution theory (Corollary 2). A low quantile of the estimated distances is only a diagnostic.

▸ Implementation

## Conditioning on network classes

When the characteristic takes finitely many values, the network partitions the switchers into classes that share the movement of $\lambda$: the connected components of the graph joining switchers with the same lagged outcome whose estimated distance is below $h_1$.

### Class-conditional estimator

For a class $C$ with $N_C$ members of ordering $B_s$ (those who quit),

$$\ell_C(\theta) = \sum_{j \in C} b_{js}\, Z'_{js}\theta - \log \sum_{S \subseteq C:\ |S| = N_C} \exp\Big(\sum_{j \in S} Z'_{js}\theta\Big), \qquad \hat{\theta}^{\mathrm{C}} = \arg\max_{\theta} \sum_{s} \sum_{C} \ell_C(\theta),$$

the conditional probability of the observed orderings given how many members switched each way: Chamberlain's conditional logit with the class in the role of the individual.

- A class with one member of each ordering contributes the pair's term of the pairwise objective: the pairwise estimator is the composite version of the class likelihood.
- **Proposition 2.** With one transition period, the feasible classes are the population classes with probability approaching one, the class likelihood is exact at every sample size for any class whose members share the same movement, and $\sqrt{n}(\hat{\theta}^{\mathrm{C}} - \theta) \to_d \mathcal{N}(0, \mathcal{I}_{\mathrm{C}}^{-1})$, with the inverse Hessian as variance. When $T = 3$, at least as efficient as the pairwise estimator (argued, not proved).

Details

# Large-sample theory: what is established

$T$ is fixed and $n \to \infty$; the network is dense, with degrees that grow with $n$.

## Consistency (Theorem 2)

With bandwidths that shrink at the rates of Assumption 4 ($nh_1^2/\log n \to \infty$, $n\,\omega_n \to \infty$) and an idiosyncratic movement of $\lambda$ whose maximum shrinks with $n$ (Assumption 3), $(\hat{\beta}, \hat{\alpha}) \to_p (\beta, \alpha)$, including with continuous types and covariates.

## Asymptotic normality (Theorem 3)

When the characteristic takes finitely many values, covariate changes have an atom at zero with gaps, and the bandwidths lie below the gaps, exact matches are selected with probability approaching one and the estimator is root-$n$ normal, centered at the truth.

- The kernel weights depend on estimated distances, which are functions of the whole network, so the score is not a standard U-statistic; consistency uses concavity and the uniform rate of Proposition 1.
- Variance: the sandwich that clusters the pair-level scores by agent, which is the plug-in estimator of Theorem 3; for the class estimator, the inverse Hessian (Proposition 2).
- With continuous types, consistency only: no limiting distribution is established.

▸ Conditions

# Finite-type designs: root-$n$ and centered at the truth

## Theorem 3 (asymptotic normality with finitely many values)

Under Assumptions 1, 2, 4(i), 5(i)–(ii), and 7 (finite support, equivalence on the support, exact dynamics, an atom at zero covariate change, bandwidths below the gaps, rank),

$$\sqrt{n}\begin{pmatrix}\hat{\beta}-\beta\\ \hat{\alpha}-\alpha\end{pmatrix} \to_d \mathcal{N}\big(0,\ 4\,\Sigma_{\text{ex}}^{-1} V_{\text{ex}}\,\Sigma_{\text{ex}}^{-1}\big), \qquad V_{\text{ex}} = \operatorname{Var}(\psi_i),$$

with $\psi_i$ the expectation of agent $i$'s exact-match pair scores over an independent partner.

- **Weights select exact matches.** With probability approaching one, the estimated weights are positive exactly on same-type pairs and within $o_p(n^{-1/2})$ of $K_1(0)K_2(0)^2$, so estimating the distances does not affect the limit.
- **Oracle equivalence.** The estimator behaves like a conditional logit on the true matches, a second-order U-process in the i.i.d. node data; the link shocks drop out.
- **No bias term.** On the exact-match event the score has mean zero, by Lemma 2 and Assumption 7(ii)–(iii). ▸ Assumption 7 ▸ Variance
- **Corollary 2.** With $T = 3$, the same holds for a random bandwidth between $c_- n^{-1/4}$ and $c_+ n^{-1/4}$ computed from $D_1$ and $y_0$, and so does Proposition 2. ▸ The rule

## Continuous designs: what is not established, and why

When types or covariates vary continuously, matches are inexact at every bandwidth. Consistency holds; the paper establishes no limiting distribution, for four reasons.

1. **Shared links.** The estimated weights of different pairs share network rows, so per-agent influence terms built from realized links are not independent across agents.
2. **First-stage error.** The codegree distance carries link noise and the sampling variation of the other agents' latent types; the second does not vanish when link noise is averaged out.
3. **Heterogeneous match mass.** The leading variance contains $\mathbb{E}[r_h(w_i)^2]/(\mathbb{E}[r_h(w_i)])^2$, which can diverge as $h_1 \to 0$ when agents' match probabilities vary at different rates, so there is no universal rate.
4. **Bias correction.** A generalized jackknife would remove bandwidth-driven bias if the exponents were known; they are not derived, so it is an algebraic device, not an inference result.

▸ Examples

### In practice

Point estimates are reported with the finite-type standard errors for comparison only, and the simulations report their coverage as a diagnostic, not as evidence for an inference theory.

# Monte Carlo design

- $T = 3$ plus an initial outcome from the model; $\beta_0 = 1$, $\alpha_0 = 0.5$, $\lambda(w) = (4w^3 - 3)/2$, fixed effect $\mu_i \sim \mathcal{N}(0, 1)$, independent of the index and absent from the link function.
- **Finite-type designs.** Four latent values $w \in \{0.15, 0.40, 0.65, 0.90\}$, equally likely, constant at $t = 0, 1$; at $t = 2$ every agent moves by the common cyclic permutation $k \mapsto k + 1$; constant at $t = 3$. So $d_1 \neq 0$ and $d_2 = 0$: Assumption 2 holds with $\mathcal{S} = \{1\}$.
- Covariate $X_{it} = \text{round}(2w_{it} + 0.5\mu_i + v_{it})$: correlated with both unobserved terms, integer valued. Links: a block matrix with distinct codegree profiles (separation gap $3.3 \times 10^{-3}$) in which neighbors' average index says little about an agent's own.

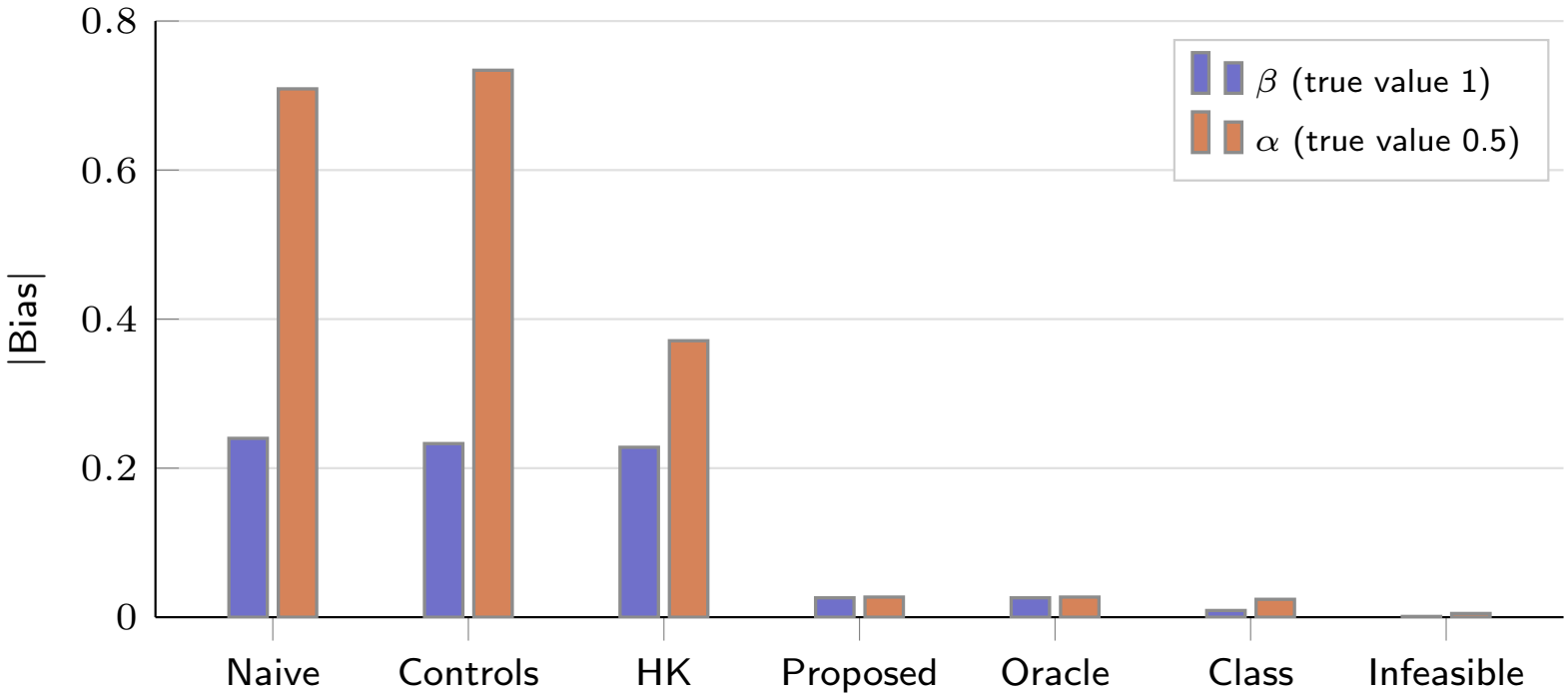
| Design | Feature |
|---|---|
| baseline | as above; the regime of Theorem 3 |
| lag-dependent | link probabilities $\times\ (1 + 0.4\, y_{i,t-1} y_{j,t-1})$: the lag stratification binds |
| fixed types | types do not move: within-agent conditioning is valid |
| weak network | block matrix shrunk toward its mean: separation gap divided by ten |
| idiosyncratic | 30% of agents draw a new type at $t = 2$: Assumption 2(i) fails |
| covariates in links | link probabilities depend on whether $X_{it} = X_{jt}$: outside the network model |
| sparse | expected degree 8 at every $n$: outside the dense-graphon asymptotics |
| continuous | $w_{i1} \sim U[0, 1]$, $w_{i2} = w_{i3} = w_{i1}^{1/2}$, steep logistic link, continuous covariate: consistency only |

# Seven estimators, ordered by what stands in for $\lambda(w_{it}) + \mu_i$

| Estimator | What replaces $\lambda(w_{it}) + \mu_i$ | Uses the network? |
|---|---|---|
| Naive | a constant | no |
| Controls | peer averages of $X$ and of lagged $y$ | as controls |
| HK | a time-invariant agent effect | no |
| Proposed | removed within (fixed effect) and across matched agents (common change) | to match agents |
| Oracle | the same, with the *true* matches | not needed |
| Infeasible | the true $\lambda(w_{it}) + \mu_i$ as a regressor | not needed |
| Class | the same comparisons, class by class | to form classes |

- In the finite-type designs except the sparse one, bandwidths are the deterministic sequences of the theory: $h_1 = 0.5\,\bar{\epsilon}\,(n/500)^{-1/4}$ below the separation gap, and $h_2 = 0.5$, which selects exact covariate stability (for HK as well). The continuous design has its own rates, and a data-driven $h_1$ comes at the end of this section.
- $n \in \{500, 1000, 2000, 3000\}$, $R = 500$ replications. A fit fails when fewer than three pairs are selected or the optimizer fails; a linear program checks each optimizer failure for separation.
- The question: how much of the gap between Naive and Infeasible can the network close?

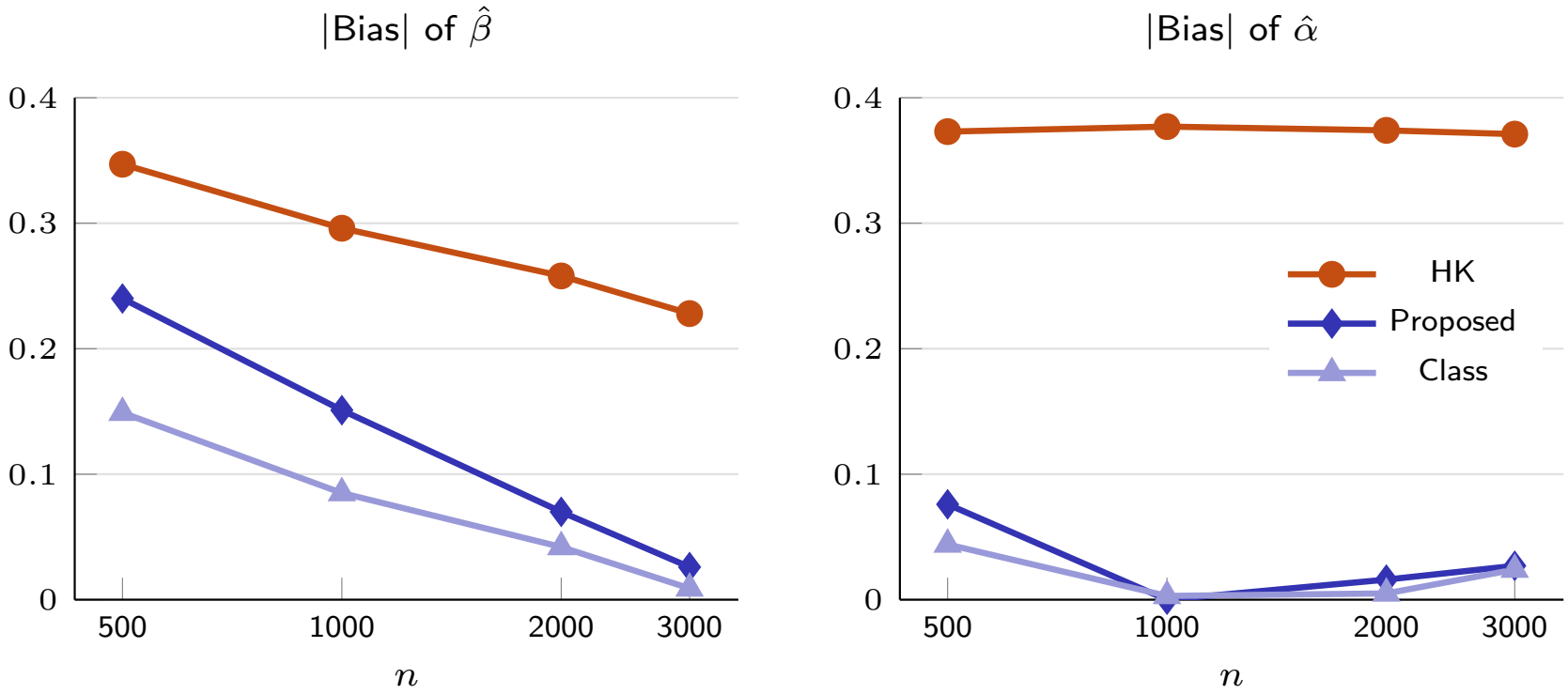
# The bias ladder at $n = 3000$, baseline design




- Pooled and peer-control logits attribute the omitted persistent heterogeneity to state dependence: bias 0.71 to 0.73 in $\alpha$.
- HK removes the fixed effect and the level of $\lambda$ but not its common movement: 0.23 for $\beta$, 0.37 for $\alpha$.
- The proposed estimator agrees with the oracle to the third decimal; the class estimator is closer still.

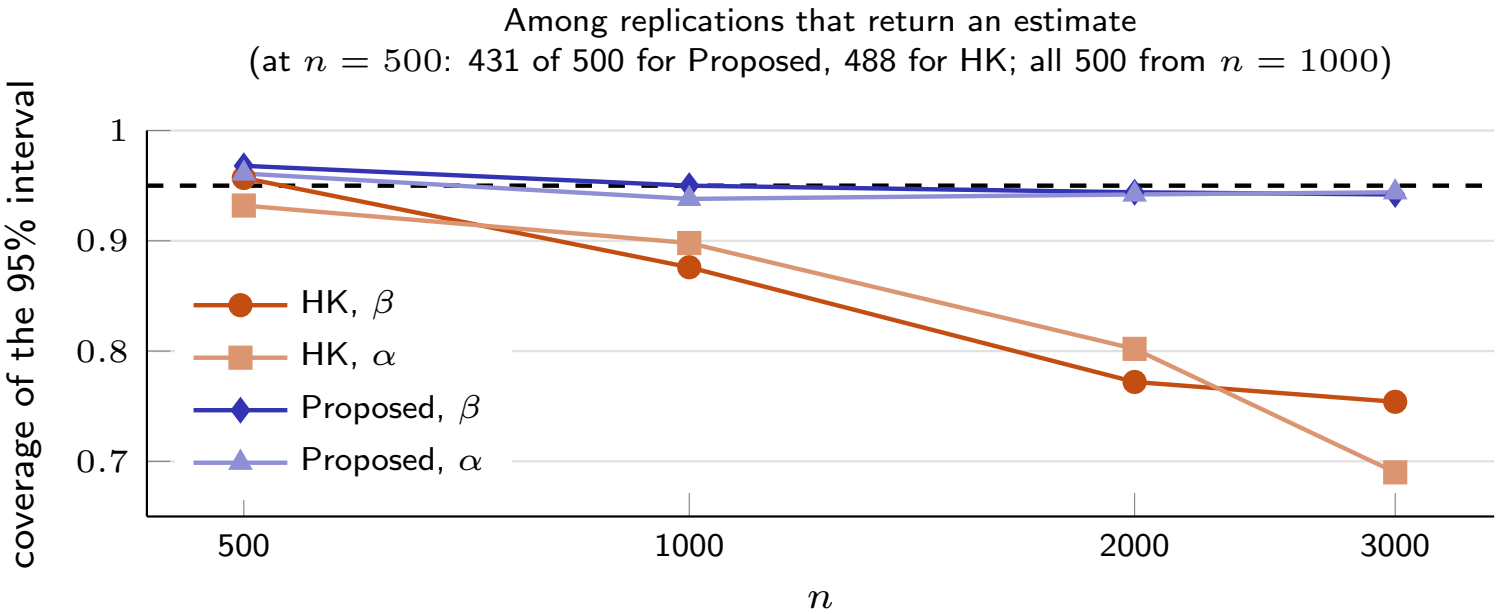
▸ Full tables

# Bias as the sample grows, baseline design




- HK's bias comes from the moving characteristic, not from small samples: it does not vanish as $n$ grows.
- The proposed estimator's bias for $\beta$ falls from 0.24 to 0.03; the bias at $n = 500$ is shared by the oracle, so it is the finite-sample bias of a conditional logit with few contributing agents, not a matching error.

# Confidence intervals




- HK's coverage falls to 0.75 for $\beta$ and 0.69 for $\alpha$ by $n = 3000$: its bias stays while its standard error shrinks. Pooled and peer-control coverage is at most 0.004 from $n = 1000$ onward.
- The proposed estimator covers in 0.94 to 0.97 of the replications that return an estimate; the ratio of the Monte Carlo standard deviation to the clustered standard error is 1.02 to 1.16, so the standard errors are slightly too small.
- At $n = 500$, 69 of 500 fits fail because the likelihood is separated; from $n = 1000$ every fit succeeds.

# The price: variance

| | RMSE of $\hat{\beta}$ | | | RMSE of $\hat{\alpha}$ | | |
|---|---|---|---|---|---|---|
| $n$ | HK | Proposed | Class | HK | Proposed | Class |
| 500 | **0.568** | 0.677 | 0.563 | **0.825** | 1.300 | 1.163 |
| 1000 | **0.433** | 0.458 | 0.381 | **0.614** | 0.872 | 0.757 |
| 2000 | 0.321 | **0.262** | 0.222 | **0.499** | 0.543 | 0.477 |
| 3000 | 0.275 | **0.201** | 0.174 | 0.457 | **0.419** | 0.380 |

Each RMSE uses that estimator's own fits (at $n = 500$: 488 for HK, 431 for Proposed, 435 for Class); on common fits no ranking changes.

- Only matched switchers with stable covariates count. At $n = 500$, about 50 agents are eligible, 36 have a partner, and the least-matched one has a single partner; at $n = 3000$, 294 of 299 contribute.
- The information grows with the number of contributing agents, roughly in proportion to $n$, not with the number of pairs, which share agents.
- At small $n$ HK has the smaller RMSE; from $n = 2000$ for $\beta$ and $n = 3000$ for $\alpha$ the ordering reverses. The inference comparison is the sharper summary.

# Where the method loses ground, and where it fails

| | HK | | | Proposed | | | Oracle | | |
|---|---|---|---|---|---|---|---|---|---|
| $\beta$, $n = 2000$ | Bias | RMSE | Cov. | Bias | RMSE | Cov. | Bias | RMSE | Cov. |
| Baseline | 0.258 | 0.321 | 0.77 | 0.070 | 0.262 | 0.94 | 0.069 | 0.262 | 0.94 |
| Fixed types | **0.030** | **0.195** | 0.94 | 0.046 | 0.223 | 0.96 | 0.046 | 0.223 | 0.96 |
| Weak network | 0.252 | 0.322 | 0.81 | 0.110 | 0.362 | 0.94 | 0.064 | 0.268 | 0.95 |
| Idiosyncratic movement | 0.245 | 0.317 | 0.79 | 0.131 | 0.287 | 0.93 | 0.131 | 0.287 | 0.93 |
| Covariates in links | 0.258 | 0.327 | 0.78 | 0.104 | 0.394 | 0.95 | 0.064 | 0.249 | 0.94 |
| Sparse network | 0.223 | 0.292 | 0.85 | 0.555 | 0.833 | 0.78 | 0.036 | 0.239 | 0.96 |

- **Fixed types:** HK is valid and more precise; the network adds nothing.
- **Weak network:** the separation gap is near the noise of the estimated distances; at $n = 1000$ no pair is selected, and recall is 0.45 at $n = 2000$ and 0.79 at $n = 3000$.
- **Idiosyncratic movement:** Assumption 2(i) fails and the bias doubles, for the oracle too.
- **Covariates in links:** matches are fewer (recall 0.32), and the RMSE rises to 0.394, against 0.249 for the oracle.
- **Sparse network:** the codegree distance carries little information when degrees stay bounded, and the matching fails; the oracle with true matches is centered.

▸ Table with $\alpha$ and the class estimator

# Conditioning on network classes: the gain

- In the baseline and lag-dependent designs, the class estimator has the smaller RMSE for both parameters at every sample size, by 6 to 20 percent, with coverage between 0.95 and 0.99.
- The feasible classes coincide with the population classes in 88 percent of replications at $n = 500$ and in at least 99.6 percent from $n = 1000$, where the class estimator's RMSE is within 0.001 of the class oracle's: the gain comes from the likelihood, not from the classification.
- **The classes matter, not just the conditioning.** A single class per lag stratum, one common movement for all agents and no network, is biased by 0.229 for $\beta$ at $n = 3000$, as HK is (0.228).
- **Conditioning matters, not just the classes.** Estimating one intercept per class jointly gives a bias of 0.485 at $n = 500$, against 0.149 for exact conditioning.
- **Continuous design.** The pairwise estimator is consistent: bias for $\beta$ from 0.43 at $n = 500$ to 0.06 at $n = 3000$, within 0.05 of the oracle, while HK stays at 0.30 to 0.44; coverage of 0.95 to 0.98 is a diagnostic only.

▸ Table ▸ Class table

## A data-driven bandwidth when $T = 3$

$\hat{h}_1 = 0.1 \max\{0.001,\ S_n\}\,(n/500)^{-1/4}$, with $S_n$ the largest first-wave squared codegree distance among pairs of the first $\lceil \sqrt{n} \rceil$ agents sharing $y_0$. It uses neither the types nor the gap, and Corollary 2 covers it.

| | | Pairwise with $\hat{h}_1$ | | | Oracle | | |
|---|---|---|---|---|---|---|---|
| $\beta$ | $n$ | Fits | Bias | RMSE | Fits | Bias | RMSE |
| Baseline | 500 | 426 | 0.256 | 0.715 | 431 | 0.235 | 0.670 |
| | 1000 | 500 | 0.152 | 0.457 | 500 | 0.151 | 0.459 |
| | 2000 | 500 | 0.070 | 0.263 | 500 | 0.069 | 0.262 |
| Weak network | 500 | 33 | −0.087 | 0.850 | 425 | 0.238 | 0.665 |
| | 1000 | 322 | 0.278 | 0.787 | 500 | 0.133 | 0.451 |
| | 2000 | 500 | 0.083 | 0.304 | 500 | 0.064 | 0.268 |

- **Baseline:** $\hat{h}_1$ is 0.11 to 0.33 times the gap, so only same-type pairs are selected. From $n = 1000$ the estimator is within 0.008 of the oracle.
- **Weak network:** the noise of the distances is of the order of the gap, so the rule finds only 28 and 39 percent of the exact matches at $n = 500$ and 1000, yet it returns 322 estimates at $n = 1000$, where the calibration returns none.
- **The constant matters in small samples:** halving it leaves no estimate in the weak design at $n \leq 1000$.

Full table

# Glasgow Teenage Friends and Lifestyle Study: a feasibility exercise

- One secondary school in Glasgow, three waves: 1995, 1996, and 1997 (Michell and West, 1996; Pearson and Michell, 2000).
- 160 pupils, 129 present at all waves, and 96 with complete data.
- **Outcome:** any recorded tobacco or cannabis use.
- **Covariates:** pocket money, romantic relationship, alcohol, leisure.
- **Network:** up to six friend nominations per wave, symmetrized by the OR rule.

| | 1995 | 1996 | 1997 |
|---|---|---|---|
| Use (tobacco or cannabis) | 0.417 | 0.521 | 0.708 |
| Pocket money (£) | 9.67 | 11.70 | 16.21 |
| Alcohol (1–5) | 2.78 | 3.06 | 3.48 |
| Density | 0.041 | 0.039 | 0.037 |
| Average degree | 3.9 | 3.7 | 3.5 |

Means in the estimation sample. ▸ Table

## Why this setting, and what the exercise is

Friendships and smoking co-evolve during adolescence, when risk attitude and attachment to norms also move. The data fall short of the model's requirements in documented ways, so the exercise tests feasibility: the conditional fits are descriptive, with no inferential claim attached.

## What three waves and 96 students allow

- **Initial condition.** Four dates are needed and three are observed, so I set $y_{i0} = 0$ for all students. They are 13 at wave 1, but 41.7% already report some use: the classical initial conditions problem (Heckman, 1981b).
- **Transitions carry the comparison.** 22 students switch between waves 1 and 2 (16 start and 6 quit), which gives $16 \times 6 = 96$ discordant pairs, all in the lag stratum $y_{i0} = 0$.
- **No covariate stability.** None of the 22 has stable covariates between waves 2 and 3, so $K_2$ is omitted, and the period-3 term $\Gamma(c_{i3}) - \Gamma(c_{i2})$ stays in every conditional fit.
- **The network does not bind unless forced.** The largest distance among the 96 pairs is $1.8 \times 10^{-4}$, so a bandwidth such as $n^{-1/5} \approx 0.40$ gives all pairs practically the same weight: the unlocalized pairwise fit does not use the network, and the class fit has a single class of 22 switchers.
- **Near-separation.** The unpenalized conditional likelihoods exist but are near-separated (half of HK's and 76% of the pairwise fitted probabilities outside $[0.01, 0.99]$), so the conditional fits are reported with Firth penalization (Firth, 1993).
- **The network observation departs from the model:** a nomination cap of six, the OR rule, density 0.04, and induced subgraphs.

▸ Network measurement

# Descriptive fits

| | (1) Naive | (2) Controls | (3) HK | (4) Pairwise | (5) Class |
|---|---|---|---|---|---|
| Lagged use ($\hat{\alpha}$) | 2.222 | 2.101 | 1.827 | 1.413 | 0.893 |
| | (0.420) | (0.444) | (0.668) | (1.368) | (1.536) |
| Alcohol | 0.808 | 0.800 | 1.501 | 3.253 | 1.698 |
| | (0.174) | (0.185) | (0.414) | (0.858) | (0.988) |
| Pocket money | 0.014 | 0.010 | 0.062 | 0.099 | 0.059 |
| | (0.019) | (0.020) | (0.046) | (0.050) | (0.064) |
| Firth penalization | — | — | yes | yes | yes |
| Observations | 288 | 288 | 22 agents | 96 pairs | 22 agents |

- The point estimate of state dependence is positive in each of the five unlocalized fits, falling from 2.22 to 0.89 as more conditioning is imposed; the pocket money coefficient is imprecise throughout, and alcohol is the covariate most strongly associated with use.
- Columns (4) and (5) do not use the network: the move from 1.83 reflects a change from within-agent to cross-agent conditioning, not the removal of the class-specific movement that matching targets.
- Standard errors are for scale only; without penalization HK's $\hat{\alpha}$ is 3.88 and the ordering breaks.

▸ Full table ▸ Robustness

## When the network binds

Bandwidth $h_1$ at the $q$-th quantile of the aggregated codegree distances, so that the codegree kernel actually selects pairs:

| | Any friend | | | Best friend |
|---|---|---|---|---|
| | $q = 0.10$ | $q = 0.20$ | $q = 0.30$ | $q = 0.10$ |
| Pairs with positive weight | 9 | 21 | 26 | 14 |
| Effective sample size | 7.0 | 17.1 | 23.0 | 14.0 |
| Max. min. signed margin ($\|b\| = 1$) | 1.540 | 0.566 | 0.483 | 0.729 |
| MLE exists | no | no | no | no |
| Firth-penalized $\hat{\alpha}$ | 0.176 | $-0.217$ | $-0.403$ | 0.876 |

- A strictly positive margin means that some direction orders every weighted pair correctly, which certifies complete separation (Albert and Anderson, 1984): the likelihood has no maximizer.
- The Firth-penalized estimates are finite but uninformative, and $\hat{\alpha}$ changes sign across bandwidths.
- The class estimator does no better: at each bandwidth a single class contains both orderings, and its likelihood is separated too.
- At $n = 96$ with a sparse capped network, the data sustain the differencing but not the codegree localization.

▸ Full table

# What the application teaches

## A division of labor

The application exercises the differencing, and the simulations exercise the matching. No part of the application validates the full procedure; it documents the data requirements of the method.

A dataset suited to the method needs:

- at least four waves, so that the initial outcome is observed rather than imputed;
- enough stable switchers within recoverable network classes, so that the covariate-stability condition can be imposed and the localized likelihood is not separated;
- a network observed without a nomination cap, with degrees large enough for codegrees to be informative.

Two features help: covariates recorded on discrete scales, which give the exact matches of the inference results, and dates close together around a scheduled transition, which make the common change and the stability after it more plausible.

Such data would address the obstacles documented here; the overlap of types among switchers, the common movement, strict exogeneity, and the rank condition would still have to be assessed, and none is a matter of sample size alone.

# Summary

- **Problem.** Persistence in a choice and the formation of friendships can be driven by the same unobserved characteristics, part of which change over time; a fixed effect removes only the part that does not.
- **Identification.** Within a student, a switch removes the fixed effect and the level of the social influence function. Across two students of the same network type who switch in opposite directions, the comparison removes its change, if the change is common within a network type and followed by a stable wave. No parametric link model is assumed, and no peer effect is identified.
- **Estimation and inference.** A pairwise conditional likelihood estimator on codegree-matched pairs: consistent, including with continuous types, and root-$n$ normal, centered at the truth, with finitely many values and exact matches, also with a data-driven bandwidth when $T = 3$. A class-conditional estimator is root-$n$ normal with one transition period and, when $T = 3$, at least as efficient (argued).
- **Evidence.** In simulations the pairwise estimator agrees with an oracle that knows the matches, its bias shrinks with $n$, and its intervals cover near the nominal rate among successful fits, while within-agent conditioning stays biased when the characteristic moves; the class estimator cuts the RMSE by 6 to 20 percent in the two main designs. The Glasgow exercise documents the data the method requires.

## Next steps

- **Continuous types.** A limiting distribution under a regularity condition on the link function, by conditioning on the period-1 network so that the estimated weights are fixed; work in progress, not part of the paper.
- **Feedback.** Outcome shocks that feed into later covariates and into the latent characteristic, which strict exogeneity now rules out.
- **Data.** A school panel with four or more waves, repeated friendship nominations without a cap, and a scheduled transition; requests for restricted-access panels are under way.
- **Networks as observed.** Directed, censored, and misreported links: separate out- and in-codegree distances, corrections for nomination caps, and a distance that remains a proxy for type similarity under misclassification.
- **Extensions in the paper.** What is identified about the social influence function ▸ Appendix B and ordered outcomes ▸ Appendix C.

Thank you

bgueyapk@uwo.ca

# Assumption 1: sampling and distributional structure

(i) $(\chi_i, \varepsilon_i) = (y_{i0}, X_i, w_i, \mu_i, \varepsilon_i)$ are i.i.d. across agents; temporal dependence within an agent is allowed, and $\mu_i$ may depend on $(y_{i0}, X_i, w_i)$ arbitrarily.

(ii) The link shocks $\{\eta_{ijt}\}$ are symmetric, i.i.d. standard uniform above the diagonal for each $t$, independent across $t$ and of $(\chi_i, \varepsilon_i)$: unobserved factors that affect both linking and outcomes operate through $w_{it}$.

(iii) For each $t$, $w_{it}$ has a distribution $G_{w,t}$ on $[0, 1]$, which may differ across periods, since the index moves.

(iv) $\varepsilon_{it}$ is standard logistic, i.i.d. across $i$ and $t$, and independent of $\chi_i$. Strict exogeneity: it rules out, for example, a risk attitude or a covariate that responds to the agent's own past smoking.

(v) Outcomes and networks are generated by the outcome and network equations.

(vi) $\lambda$ and $f$ are measurable, with $f$ jointly symmetric: $f(w_1, w_2; y, y') = f(w_2, w_1; y', y)$.

The logistic form is what makes the conditional likelihood work: the cancellation relies on the log-linear odds ratio (Andersen, 1970; Chamberlain, 1980), which is essentially unique to the logistic family (Magnac, 2004). Without it the problem becomes one of partial identification, which the paper does not pursue.

Back

# Theorem 1: conditioning events and proof

Fix an interior period $s$ and two agents $i \neq j$. Let $\mathcal{F}_{ijs} = \sigma(X_i, X_j, y_{i,-s}, y_{j,-s})$ be the covariates and the outcomes outside the window, $\mathcal{D}_{ijs}$ the discordance event, and

$$\mathcal{C}_{ijs} = \{\lambda(w_{is}) - \lambda(w_{i,s+1}) = \lambda(w_{js}) - \lambda(w_{j,s+1});\ \lambda(w_{a,s+2}) = \lambda(w_{a,s+1}),\ a \in \{i,j\}\},$$
$$\mathcal{X}_{ijs} = \{X_{a,s+2} = X_{a,s+1},\ a \in \{i,j\}\}, \qquad \mathcal{O}_{ijs} = \{y_{i,s-1} = y_{j,s-1},\ \delta_{ijs} = 0\} \cap \mathcal{X}_{ijs},$$
$$Z_{ijs} = \big(\Delta_s X_i - \Delta_s X_j,\ y_{i,s-1} - y_{i,s+2} - y_{j,s-1} + y_{j,s+2}\big)'.$$

**Lemma 2 (conditional likelihood identity).** On $\mathcal{C}_{ijs} \cap \mathcal{X}_{ijs}$, given $\mathcal{F}_{ijs}$, $(w_i, w_j, \mu_i, \mu_j)$, and $\mathcal{D}_{ijs}$, the probability of $A_s \times B_s$ is $F(V_{ijs}(\beta, \alpha))$ with $V_{ijs} = -Z'_{ijs}\theta$. **Proof.** Agents are independent given their primitives, so the joint path probabilities factorize; Lemma 1 applied to both agents removes both fixed effects, the equal changes cancel, and given discordance only two configurations remain.

**Theorem 1(a).** If the same type implies the same change (finite support with equivalence on the support, or continuity of $f$ and full support), then $\mathcal{O}_{ijs} \subseteq \mathcal{C}_{ijs} \cap \mathcal{X}_{ijs}$ and the identity holds given the observable event; the fixed effects never enter. With $\mathbb{E}[Z_{ijs}Z'_{ijs} \mid \mathcal{O}_{ijs}, \mathcal{D}_{ijs}]$ positive definite, $(\beta, \alpha)$ is the unique maximizer of the conditional expected log-likelihood.

**Theorem 1(b).** With continuous types, on the neighborhood $\mathcal{O}_{ijs}(h)$ the identity holds up to a remainder bounded by $\varpi(\tau(h)) + 2\|\beta\|h^{1/2} \to 0$; the localized criterion has a unique maximizer $\theta_h \to (\beta, \alpha)$.

The rank condition is imposed on the selected population, with the shared lag: on it, the second component of $Z_{ijs}$ reduces to $y_{j,s+2} - y_{i,s+2}$.

Back

# Why four dates are needed

**Lemma 3 (minimum number of periods).** The strategy uses three consecutive periods $(s, s+1, s+2)$ plus the lag at $s-1$: at least four dates. With $T = 2$ and $\alpha \neq 0$, $(\beta, \alpha)$ is not identified *by this strategy*.

With $c_{i1} = X_{i1}'\beta + \alpha y_{i0} + \lambda(w_{i1}) + \mu_i$ and $c_{i2} = X_{i2}'\beta + \lambda(w_{i2}) + \mu_i$,

$$\mathbb{P}\big(y_i \in A_1 \mid y_{i1} + y_{i2} = 1, \chi_i\big) = \frac{e^{c_{i2}}\left(1 + e^{c_{i2}+\alpha}\right)}{e^{c_{i2}}\left(1 + e^{c_{i2}+\alpha}\right) + e^{c_{i1}}\left(1 + e^{c_{i2}}\right)}.$$

- The factor $(1 + e^{c_{i2}+\alpha})/(1 + e^{c_{i2}})$ depends on $\lambda(w_{i2})$ and $\mu_i$ in a way that is not log-linear, so no cross-agent comparison removes it.
- With a third period, period $s+2$ contributes the reciprocal factor under the stability condition and cancels it exactly; this is where $y_{i3}$ and the stability of covariates and of $\lambda$ over wave 3 enter.
- When $\alpha = 0$ is known, the factor equals one and $\beta$ is identified at $T = 2$ by the cross-agent comparison alone: the impossibility is specific to the dynamic parameter.

Back

# Population codegree distance and Proposition 1

Conditional codegree function, with $G_t$ the joint distribution of $(w_{lt}, y_{l,t-1})$:

$$p(w_{it}, w_{jt};\, y_{i,t-1}, y_{j,t-1}) = \int f(w_{it}, u;\, y_{i,t-1}, y)\, f(w_{jt}, u;\, y_{j,t-1}, y)\, dG_t(u, y),$$

the expected fraction of common neighbors. For agents with $y_{i,t-1} = y_{j,t-1}$,

$$\delta_{ijt} = \Big[ \int \left( p_{w_{it};\, y_{i,t-1}, y'}(s) - p_{w_{jt};\, y_{j,t-1}, y'}(s) \right)^2 dG_t(s, y') \Big]^{1/2}.$$

**Proposition 1.**

- (a) Uniformly over pairs sharing the lag, $\max |\hat{\delta}^2_{ijs} - \delta^2_{ijs}| = O_p\big((\log n/n)^{1/2}\big)$: $\delta_{ijs}$ is identified in the growing-network experiment.
- (b) Under continuity of $f$ and full support (Assumption 6), $\delta_{ijs} = 0$ if and only if the two agents have the same network type, and $\delta^2_{ijs} \leq h$ implies a small difference in the common movement, $|d_s(w_{is}) - d_s(w_{js})| \leq \varpi(\tau(h))$.
- (c) When the characteristic takes finitely many values, $\delta_{ijs} = 0$ if and only if the agents are equivalent on the support, and every positive value of $\delta^2_{ijs}$ is at least $\bar{\epsilon} > 0$.

Identification is understood relative to the growing-network experiment, as in Auerbach (2022): a pair-level quantity is identified when it is the uniform probability limit of a statistic of the observed data.

◂ Back

# Assumption 2: common maps, what is excluded, and the choice of $\mathcal{S}$

- **Common map.** If $w_{i,t+1} = g_t(w_{it})$, then $d_t = \lambda \circ g_t - \lambda$, and part (i) holds when this induced change is constant on each network-type class and uniformly continuous; movements of any size are then allowed.
- **A class-preserving map does not suffice.** With the lag-independent link function $f(u,v) = \frac{1}{4} + \frac{1}{4}\left[(u-\frac{1}{2})^2 + (v-\frac{1}{2})^2\right]$, the indices $u$ and $1-u$ have identical link functions. The map $g(u) = 1-u$ sends each index to its own class, yet with $\lambda(u) = u$ the change $d(u) = 1-2u$ equals $0.6$ at $u = 0.2$ and $-0.6$ at $u = 0.8$: two equivalent indices move by different amounts.
- **Examples that satisfy it.** A cohort-wide change in risk attitude between two waves whose size depends only on the student's position in the friendship structure; the simulation designs, where every class is a single type and types move by a common permutation.
- **Part (ii) and the choice of $\mathcal{S}$.** If every interior period is used, $d_2 = \cdots = d_{T-1} = 0$, so the only unrestricted movement is between periods 1 and 2; with longer panels one can alternate windows of movement and stability. With $T = 3$ and $\mathcal{S} = \{1\}$, the restriction is $d_2 \equiv 0$.
- **Excluded.** Movement that differs by a non-vanishing amount across agents of the same type, such as a stationary autoregression with nondegenerate innovations; neither within- nor cross-agent comparisons remove it. The asymptotic theory allows an idiosyncratic component whose maximum shrinks with $n$ (Assumption 3).

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# The observation window in the smoking example

Four dates $t = 0, 1, 2, 3$ at comparable intervals around a scheduled school transition between $t = 1$ and $t = 2$: the outcome at $t = 0$ fixes the lag stratum, the network at $t = 1$ carries each student's class, and the outcomes at $t = 2, 3$ are observed after the transition, within the same stage. A special case of Assumption 2 is $\lambda(w_{i1}) = \ell_{c(i)}$, $\lambda(w_{i2}) = \lambda(w_{i3}) = \ell_{c(i)} + \Delta_{c(i)}$, with persistent differences within a class carried by $\mu_i$.

- **With the model, three substantive parts.** Students who formed friendships in the same way before the transition share the change in their pull toward smoking; the adjustment is complete by $t = 2$ and the latent term does not move between $t = 2$ and $t = 3$; the state-dependence parameter is the same across intervals, which requires comparable spacing.
- The realized network can still change within the window, because links depend on lagged smoking and link shocks change: stability concerns the latent contribution to the outcome, not every friendship.
- A scheduled transition does not by itself deliver these conditions: continued adjustment after $t = 2$, a latent propensity that responds to recent smoking, and a direct effect of friends' behavior are all outside the model.
- If every $\Delta_c$ is zero, within-agent conditioning is valid and the network is not needed; if the increments differ across classes, a common time effect does not remove them.
- **Glasgow.** The waves are a year apart, in the second, third, and fourth years of secondary school; the third and fourth years were those of the Standard Grade courses, so the first two waves straddle a change of stage and the last two lie within one. The data fall short: $y_{i0}$ is not observed, and stability over a full year is a strong requirement.

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# What is identified about the social influence function

**The level is not identified.** Replacing $(\lambda, \mu_i)$ by $(\lambda + c, \mu_i - c)$ leaves every outcome probability unchanged; when $w_{it} = w_i$, only $\lambda(w_i) + \mu_i$ enters. The conditional choice probability given the covariates, the lag, and the network type is an average of $F$ over the conditional distribution of $\mu_i$, so inverting it does not return $\lambda(w_{it})$.

**The common movement is identified (Corollary 1).** Given $(\beta, \alpha)$, a transition period $s$, a lag stratum, and a reference agent $i$, $d_s(w_{is})$ is the unique solution $d$ of

$$\mathbb{E}\big[\mathbb{1}\{y_j \in B_s\} - F(Z'_{js}\theta - d) \,\big|\, j \text{ in } i\text{'s class},\ j \text{ switches at } s \text{ with stable covariates},\ \chi_i\big] = 0,$$

the log odds of switching down rather than up among the switchers of the class; with continuous types, the solution on codegree neighborhoods converges to it.

**Estimator (Appendix B).** A kernel-weighted logit with a single free parameter, with the codegree kernel in place of class membership and the covariate kernel in place of exact stability, given the first-step $\hat{\theta}$; when the characteristic takes finitely many values it solves the score equation for the intercept of a conditional logit within the agent's own cell. No asymptotic theory is claimed for it, and the Glasgow data do not support it.

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# Implementation choices

- **Codegree bandwidth.** Theorems 2 and 3 use deterministic sequences. The main simulations calibrate them from the population separation gap, $h_1 = 0.5\,\bar{\epsilon}\,(n/500)^{-1/4}$, which is infeasible in practice. With $T = 3$, the rule $\hat{h}_1 = 0.1\max\{0.001, S_n\}(n/500)^{-1/4}$ depends only on $D_1$ and $y_0$, and Corollary 2 keeps the theory with it.

- **Scale.** The estimated distances concentrate around population values whose scale is design dependent, so a fixed function of $n$ can exceed the largest distance, and every pair then receives weight. A low quantile of the finite distances is a scale-adaptive diagnostic, not part of the theory.

- **Diagnostics to report.** The number of pairs with positive weight, the kernel-weighted effective sample size $(\sum_p w_p)^2/\sum_p w_p^2$, the number of eligible agents, the number of contributing agents, the number left unmatched, and the smallest match count among contributing agents.

- **Kernel.** The simulations and the application use $K(x) = \frac{3}{4}(1-x^2)\mathbb{1}(0 \le x < 1)$.

- **Standard errors.** The sandwich that clusters the pair-level scores by agent; for the class estimator, the inverse Hessian of the class-conditional log likelihood.

- **Separation.** With few weighted pairs the conditional likelihood can be completely separated. I certify separation by the largest minimum signed margin over unit-norm directions (a linear program) and report Firth-penalized estimates only as finite penalized fits.

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# Kernel, bandwidth, and regularity conditions

## Assumption 3 (drifting idiosyncratic movement)

A triangular array under which $\lambda(w_{i,t+1}) - \lambda(w_{it}) = d_t(w_{it}) + e_{it,n}$ with $|e_{it,n}| \leq \zeta_n \to 0$, the functions $d_t$ as in Assumption 2, and the distribution of $(w_{it}, y_{i,t-1})$ converging weakly.

## Assumption 4 (kernels and bandwidths)

$K_1, K_2$ non-negative, Lipschitz, with support $[0,1)$ and $K_\ell(0) > 0$; $h_1, h_2 \to 0$, $nh_1^2/\log n \to \infty$, and $n\,\omega_n \to \infty$, with $\omega_n$ the expected weight of a pair.

## Assumption 5 (regularity)

Bounded fourth moments of $X_{it}$; a compact convex parameter space with $\theta$ interior; bounded moments of the regressor within the matching windows; and a rank condition on the *selected* population, $\liminf_n \lambda_{\min}(\Sigma_n) > 0$.

## Assumption 6 (separation of network types)

$f$ is continuous and not constant, and $(w_{it}, y_{i,t-1})$ has full support on $[0,1] \times \{0,1\}$.

The condition $nh_1^2/\log n \to \infty$ keeps the uniform error of the estimated distances small relative to the window; $n\omega_n \to \infty$ requires the expected number of effective partners of an agent to grow.

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# Finite-type designs: Assumption 7

(i) For each $t$, $w_{it}$ takes finitely many values, each with positive probability.

(ii) **Equivalence on the support.** Supported indices that are equivalent on the support at $t$ in stratum $y$ have the same $d_t$.

(iii) **Exact dynamics.** The law of the primitives does not depend on $n$: no idiosyncratic movement.

(iv) $\mathbb{P}(X_{i,s+2} = X_{i,s+1}) > 0$, and $\mathbb{P}(0 < \|X_{i,s+2} - X_{i,s+1}\|^2 < \bar{\epsilon}_X) = 0$: an atom at zero with a gap.

(v) **Bandwidths.** $h_2 \leq \bar{\epsilon}_X$ fixed; $h_1 \leq \bar{\epsilon}/(2(T-2))$, with $\bar{\epsilon}$ the smallest positive value of $\delta_t^2$ over supported pairs; and $\sqrt{n}\, h_1 / \log n \to \infty$.

(vi) **Rank.** $\Sigma_{\text{ex}} = \sum_s \mathbb{E}[\mathbb{1}_{ijs}^{\text{ex}} F(Z'_{ijs}\theta)(1 - F(Z'_{ijs}\theta)) Z_{ijs} Z'_{ijs}]$ is positive definite.

**What the codegree data certify, and what must be assumed.**

- The period-$s$ equality of types is certified, because the pair shares the lag at $s-1$ and period $s$ enters the distance.
- The equal change between $s$ and $s+1$ comes from Assumption 7(ii); periods $s+1$ and $s+2$ never enter the distance of a discordant pair, and stability after the transition concerns a single agent's path.

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# Variance estimation

With $\hat{W}_{ijs}$ the estimated weight of pair $(i,j)$ at $s$, $\bar{K} = K_1(0)K_2(0)^2$, and $S_{ijs}$ the pair score,

$$\hat{\Sigma} = \binom{n}{2}^{-1} \sum_s \sum_{i<j} \frac{\hat{W}_{ijs}}{\bar{K}} \mathbb{1}_{ijs} F(Z'_{ijs}\hat{\theta})\big(1 - F(Z'_{ijs}\hat{\theta})\big) Z_{ijs} Z'_{ijs},$$

$$\hat{\psi}_i = \frac{1}{n-1} \sum_{j \neq i} \sum_s \frac{\hat{W}_{ijs}}{\bar{K}} \mathbb{1}_{ijs} S_{ijs}(\hat{\theta}), \qquad \hat{V} = \text{sample variance of } \hat{\psi}_i,$$

and $4\hat{\Sigma}^{-1}\hat{V}\hat{\Sigma}^{-1}/n$ estimates the variance of $\hat{\theta}$.

- It coincides, up to finite-sample normalization, with the sandwich that clusters the pair-level scores by agent: pairs that share an agent are dependent, and the clustering accounts for it.
- Consistency follows from the oracle equivalence in the proof of Theorem 3 and the law of large numbers for U-statistics.
- In the baseline design the ratio of the Monte Carlo standard deviation to the mean clustered standard error is 1.02 to 1.16, so the standard errors are slightly too small.
- With continuous types the same standard errors are reported for comparison only; no result justifies them there.

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# Class-conditional estimator: details

- **Feasible classes.** Among eligible agents (switchers with a positive covariate-stability weight) with the same $y_{i,s-1}$, join $i$ and $j$ when $\hat{\delta}^2_{ijs} < h_1$; the classes are the connected components. A class whose members all have the same ordering contributes nothing.
- **Proposition 2(a).** With probability approaching one, the feasible classes are the population classes $(y_{j,s-1}, [w_{js}])$: the distance is estimated at rate $(\log n/n)^{1/2}$, positive values are at least $\bar{\epsilon} \geq 2h_1$, and equivalence is transitive.
- **Proposition 2(b).** Given the primitives, the outcomes outside the window, the earlier networks, and the switch events, the orderings are independent logits with probability $F(Z'_{js}\theta - d_s(w_{js}))$; for any class determined by this information whose members share $d_s$, the class likelihood is the exact conditional probability of the orderings given $N_C$, at every sample size.
- **Proposition 2(c).** $\sqrt{n}(\hat{\theta}^{\mathrm{C}} - \theta) \to_d \mathcal{N}(0, \mathcal{I}_{\mathrm{C}}^{-1})$, with $\mathcal{I}_{\mathrm{C}}$ the information of a logit with one free intercept per class after profiling; the inverse Hessian estimates it, and with one transition period no clustering is needed.
- **Efficiency when $T = 3$ (argued, not proved).** In the model where orderings are independent logits with one intercept per class, $\mathcal{I}_{\mathrm{C}}$ is the efficient information; the pairwise estimator is regular with influence function $2\Sigma_{\mathrm{ex}}^{-1}\psi_j$, so by the convolution theorem its variance is at least $\mathcal{I}_{\mathrm{C}}^{-1}$. The simulations measure the gap.
- **Computation.** The elementary symmetric polynomial by the recursion of Gail, Lubin, and Rubinstein (1981), at cost $|C|N_C$.

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# Continuous designs: the obstacles in examples

- **First-stage error.** With $f(u,v) = uv$ and $w$ uniform, the population distance is $\delta_{ij}^2 = (w_i - w_j)^2 m_2^3$ with $m_2 = \mathbb{E}[w^2]$. Replacing links by their probabilities in the statistic gives $(w_i - w_j)^2 M_{2n}^3$ with $M_{2n} = n^{-1}\sum_l w_l^2$, whose fluctuation $3(w_i - w_j)^2 m_2^2 (M_{2n} - m_2)$ is of order $n^{-1/2}$ and does not vanish when link noise is averaged out.

- **Heterogeneous match mass.** With $T = 3$, $w_{it} = w_i$ uniform, $\lambda = 0$, $\mu_i = 0$, $\beta = \alpha = 0$, and the smooth, separating link $f(u,v) = \frac{1}{4} + \frac{1}{4}(u^4 + v^4)$, the squared distance is proportional to $(u^4 - v^4)^2$; the match mass behaves like $b^{1/4}$ for $w \lesssim b^{1/4}$ and like $b/w^3$ otherwise, with $b = h_1^{1/2}$, so the variance factor $\mathbb{E}[r_h(w_i)^2]/(\mathbb{E}[r_h(w_i)])^2$ is of order $h_1^{-1/8}$ and the leading variance is of order $1/(nh_1^{1/8})$, not $1/n$.

- **Consequence.** There is no universal convergence rate, and a valid theory needs a variance normalization that accommodates heterogeneous match masses, possibly after equalizing match counts (Auerbach, 2021).

- **Jackknife.** The combination $\bar{\theta} = \sum_l a_l \hat{\theta}_{c_l}$ with $\sum_l a_l = 1$ and $\sum_l a_l c_l^{e_m} = 0$ would remove listed bias terms if the exponents were known; they are not derived for any continuous design.

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# Monte Carlo results: baseline design, $\beta$ ($\beta_0 = 1$)

| $n$ | Naive | Controls | HK | Proposed | Oracle | Class | Class orc. | Infeas. |
|---|---|---|---|---|---|---|---|---|
| *Bias* | | | | | | | | |
| 500 | 0.247 | 0.242 | 0.347 | 0.240 | 0.235 | 0.149 | 0.128 | 0.009 |
| 1000 | 0.245 | 0.239 | 0.296 | 0.151 | 0.151 | 0.085 | 0.084 | 0.008 |
| 2000 | 0.240 | 0.233 | 0.258 | 0.070 | 0.069 | 0.042 | 0.042 | 0.001 |
| 3000 | 0.240 | 0.233 | 0.228 | 0.026 | 0.026 | 0.009 | 0.009 | 0.001 |
| *Root mean squared error* | | | | | | | | |
| 500 | 0.256 | 0.252 | 0.568 | 0.677 | 0.670 | 0.563 | 0.548 | 0.076 |
| 1000 | 0.251 | 0.245 | 0.433 | 0.458 | 0.459 | 0.381 | 0.381 | 0.056 |
| 2000 | 0.242 | 0.236 | 0.321 | 0.262 | 0.262 | 0.222 | 0.222 | 0.036 |
| 3000 | 0.242 | 0.235 | 0.275 | 0.201 | 0.201 | 0.174 | 0.174 | 0.033 |
| *Coverage of the nominal 95% interval* | | | | | | | | |
| 500 | 0.050 | 0.058 | 0.957 | 0.968 | 0.968 | 0.963 | 0.963 | 0.934 |
| 1000 | 0.000 | 0.004 | 0.876 | 0.950 | 0.950 | 0.974 | 0.974 | 0.944 |
| 2000 | 0.000 | 0.000 | 0.772 | 0.944 | 0.944 | 0.960 | 0.960 | 0.970 |
| 3000 | 0.000 | 0.000 | 0.754 | 0.942 | 0.944 | 0.950 | 0.950 | 0.936 |
| *Replications without an estimate* | | | | | | | | |
| 500 | 0 | 0 | 12 | 69 | 69 | 65 | 69 | 0 |
| 1000–3000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

$R = 500$. Bandwidths $h_1 = 0.5\,\bar{\epsilon}\,(n/500)^{-1/4}$ and exact covariate stability. Standard errors: MLE for the pooled logits, a sandwich for HK, the agent-clustered sandwich for the pairwise estimators, the inverse Hessian for the class estimators. Statistics over the replications with an estimate; the 69 failures at $n = 500$ are all certified as separated likelihoods. ▸ $\alpha$

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# Monte Carlo results: baseline design, $\alpha$ ($\alpha_0 = 0.5$)

| $n$ | Naive | Controls | HK | Proposed | Oracle | Class | Class orc. | Infeas. |
|---|---|---|---|---|---|---|---|---|
| *Bias* | | | | | | | | |
| 500 | 0.701 | 0.723 | 0.373 | −0.076 | −0.075 | −0.044 | −0.050 | −0.016 |
| 1000 | 0.713 | 0.737 | 0.377 | 0.000 | 0.000 | 0.003 | 0.002 | 0.004 |
| 2000 | 0.708 | 0.732 | 0.374 | 0.016 | 0.016 | −0.005 | −0.005 | −0.004 |
| 3000 | 0.709 | 0.734 | 0.371 | 0.027 | 0.027 | 0.024 | 0.024 | −0.005 |
| *Root mean squared error* | | | | | | | | |
| 500 | 0.715 | 0.737 | 0.825 | 1.300 | 1.297 | 1.163 | 1.131 | 0.152 |
| 1000 | 0.720 | 0.744 | 0.614 | 0.872 | 0.872 | 0.757 | 0.757 | 0.108 |
| 2000 | 0.712 | 0.736 | 0.499 | 0.543 | 0.542 | 0.477 | 0.477 | 0.073 |
| 3000 | 0.712 | 0.736 | 0.457 | 0.419 | 0.419 | 0.380 | 0.380 | 0.061 |
| *Coverage of the nominal 95% interval* | | | | | | | | |
| 500 | 0.002 | 0.000 | 0.932 | 0.961 | 0.961 | 0.991 | 0.991 | 0.938 |
| 1000 | 0.000 | 0.000 | 0.898 | 0.938 | 0.940 | 0.964 | 0.964 | 0.932 |
| 2000 | 0.000 | 0.000 | 0.802 | 0.942 | 0.942 | 0.952 | 0.952 | 0.946 |
| 3000 | 0.000 | 0.000 | 0.690 | 0.944 | 0.944 | 0.954 | 0.954 | 0.956 |

Same design, estimators, and bandwidths as the previous table.

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# Benchmark designs at $n = 2000$: bias, RMSE, and coverage

| | HK | | | Proposed | | | Oracle | | | Class | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Design | Bias | RMSE | Cov. | Bias | RMSE | Cov. | Bias | RMSE | Cov. | Bias | RMSE | Cov. |
| $\beta$ ($\beta_0 = 1$) | | | | | | | | | | | | |
| Baseline | 0.258 | 0.321 | 0.77 | 0.070 | 0.262 | 0.94 | 0.069 | 0.262 | 0.94 | 0.042 | 0.222 | 0.96 |
| Lag-dependent links | 0.264 | 0.331 | 0.81 | 0.068 | 0.270 | 0.95 | 0.068 | 0.271 | 0.95 | 0.047 | 0.232 | 0.96 |
| Fixed types | 0.030 | 0.195 | 0.94 | 0.046 | 0.223 | 0.96 | 0.046 | 0.223 | 0.96 | 0.032 | 0.207 | 0.96 |
| Weak network | 0.252 | 0.322 | 0.81 | 0.110 | 0.362 | 0.94 | 0.064 | 0.268 | 0.95 | 0.041 | 0.246 | 0.94 |
| Idiosyncratic movement | 0.245 | 0.317 | 0.79 | 0.131 | 0.287 | 0.93 | 0.131 | 0.287 | 0.93 | 0.107 | 0.253 | 0.94 |
| Covariates in links | 0.258 | 0.327 | 0.78 | 0.104 | 0.394 | 0.95 | 0.064 | 0.249 | 0.94 | 0.051 | 0.288 | 0.95 |
| Sparse network | 0.223 | 0.292 | 0.85 | 0.555 | 0.833 | 0.78 | 0.036 | 0.239 | 0.96 | 0.312 | 0.431 | 0.90 |
| $\alpha$ ($\alpha_0 = 0.5$) | | | | | | | | | | | | |
| Baseline | 0.374 | 0.499 | 0.80 | 0.016 | 0.543 | 0.94 | 0.016 | 0.542 | 0.94 | −0.005 | 0.477 | 0.95 |
| Lag-dependent links | 0.400 | 0.508 | 0.79 | 0.000 | 0.550 | 0.94 | 0.000 | 0.550 | 0.94 | 0.001 | 0.472 | 0.95 |
| Fixed types | 0.011 | 0.321 | 0.93 | 0.023 | 0.454 | 0.95 | 0.023 | 0.454 | 0.95 | 0.023 | 0.433 | 0.95 |
| Weak network | 0.359 | 0.489 | 0.81 | −0.058 | 0.721 | 0.93 | −0.026 | 0.568 | 0.93 | −0.003 | 0.522 | 0.93 |
| Idiosyncratic movement | 0.352 | 0.482 | 0.79 | 0.122 | 0.534 | 0.95 | 0.122 | 0.533 | 0.95 | 0.144 | 0.493 | 0.94 |
| Covariates in links | 0.364 | 0.490 | 0.80 | −0.022 | 0.641 | 0.96 | −0.012 | 0.541 | 0.94 | −0.015 | 0.516 | 0.95 |
| Sparse network | 0.395 | 0.494 | 0.79 | 0.487 | 1.195 | 0.87 | 0.029 | 0.501 | 0.96 | 0.404 | 0.735 | 0.91 |

$R = 500$. The sparse design uses the 0.10 quantile of the estimated distances as $h_1$; all other finite-type designs use $h_1 = 0.5\,\bar{\epsilon}\,(n/500)^{-1/4}$. The covariate-link and sparse designs violate the maintained network model, and the idiosyncratic design violates Assumption 2(i).

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# Pairwise and class-conditional estimators: baseline and lag-dependent designs

| | | RMSE for $\beta$ | | | RMSE for $\alpha$ | | | Class coverage | | Classes | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Design | $n$ | Prop. | Class | Class orc. | Prop. | Class | Class orc. | $\beta$ | $\alpha$ | Exact (%) | Used |
| Baseline | 500 | 0.677 | 0.551 | 0.548 | 1.300 | 1.136 | 1.131 | 0.963 | 0.991 | 88.2 | 36 |
| | 1000 | 0.458 | 0.381 | 0.381 | 0.872 | 0.757 | 0.757 | 0.974 | 0.964 | 99.6 | 86 |
| | 2000 | 0.262 | 0.222 | 0.222 | 0.543 | 0.477 | 0.477 | 0.960 | 0.952 | 100.0 | 191 |
| | 3000 | 0.201 | 0.174 | 0.174 | 0.419 | 0.380 | 0.380 | 0.950 | 0.954 | 100.0 | 294 |
| Lag-dependent links | 500 | 0.674 | 0.541 | 0.540 | 1.426 | 1.206 | 1.225 | 0.984 | 0.984 | 93.4 | 36 |
| | 1000 | 0.435 | 0.354 | 0.354 | 0.917 | 0.791 | 0.791 | 0.968 | 0.952 | 99.6 | 86 |
| | 2000 | 0.270 | 0.232 | 0.232 | 0.550 | 0.472 | 0.472 | 0.960 | 0.950 | 100.0 | 190 |
| | 3000 | 0.191 | 0.172 | 0.172 | 0.415 | 0.391 | 0.391 | 0.962 | 0.950 | 100.0 | 295 |

- RMSE on the replications in which the pairwise, class, and class-oracle estimators all succeed. Exact: share of replications in which the feasible classes coincide with the population classes; Used: eligible agents in classes that contain both orderings.
- Median computing time per replication: under 2 ms for the pairwise step, 6 to 35 ms for the class step; the shared codegree distances take 0.7 to 100 ms between $n = 500$ and $n = 3000$.

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# Monte Carlo results: continuous design

| | $\beta$ ($\beta_0 = 1$) | | | | | | $\alpha$ ($\alpha_0 = 0.5$) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $n$ | Naive | HK | Prop. | Oracle | Class | Infeas. | Naive | HK | Prop. | Oracle | Class | Infeas. |
| *Bias* | | | | | | | | | | | | |
| 500 | 0.973 | 0.437 | 0.431 | 0.392 | 0.226 | 0.011 | 0.667 | 0.157 | 0.145 | 0.148 | 0.067 | −0.008 |
| 1000 | 0.960 | 0.325 | 0.111 | 0.105 | 0.078 | −0.003 | 0.670 | 0.142 | 0.052 | 0.052 | 0.025 | 0.002 |
| 2000 | 0.957 | 0.299 | 0.032 | 0.032 | 0.049 | −0.007 | 0.663 | 0.105 | 0.013 | 0.013 | 0.003 | −0.006 |
| 3000 | 0.961 | 0.321 | 0.057 | 0.056 | 0.087 | 0.001 | 0.665 | 0.094 | 0.007 | 0.008 | 0.013 | −0.003 |
| *Root mean squared error* | | | | | | | | | | | | |
| 500 | 0.985 | 0.947 | 1.808 | 1.655 | 1.108 | 0.178 | 0.678 | 0.497 | 1.176 | 1.131 | 0.795 | 0.131 |
| 1000 | 0.966 | 0.620 | 0.768 | 0.778 | 0.581 | 0.125 | 0.676 | 0.362 | 0.619 | 0.609 | 0.473 | 0.094 |
| 2000 | 0.959 | 0.473 | 0.474 | 0.470 | 0.367 | 0.085 | 0.666 | 0.264 | 0.396 | 0.401 | 0.315 | 0.066 |
| 3000 | 0.963 | 0.450 | 0.378 | 0.381 | 0.330 | 0.064 | 0.667 | 0.219 | 0.315 | 0.313 | 0.249 | 0.054 |
| *Coverage of the nominal 95% interval (diagnostic only)* | | | | | | | | | | | | |
| 500 | 0.000 | 0.922 | 0.960 | 0.947 | 0.966 | 0.944 | 0.000 | 0.960 | 0.964 | 0.964 | 0.966 | 0.974 |
| 1000 | 0.000 | 0.922 | 0.964 | 0.958 | 0.954 | 0.948 | 0.000 | 0.940 | 0.978 | 0.966 | 0.956 | 0.960 |
| 2000 | 0.000 | 0.892 | 0.968 | 0.964 | 0.964 | 0.958 | 0.000 | 0.938 | 0.954 | 0.944 | 0.950 | 0.954 |
| 3000 | 0.000 | 0.818 | 0.958 | 0.956 | 0.944 | 0.966 | 0.000 | 0.942 | 0.952 | 0.956 | 0.952 | 0.968 |

$R = 500$; $w_{i1} \sim U[0,1]$, $w_{i2} = w_{i3} = w_{i1}^{1/2}$, link $f(u,v) = F(8(u+v-1))$, continuous covariate; $h_1 = c\,(n/500)^{-1/5}$ with $c$ the population 0.10 quantile of the squared distance, $h_2 = n^{-1/5}/5$. The oracle uses the population distances computed by quadrature. The class estimator's connected components are not classes of equal movement here, and its gain over the pairwise estimator comes from pooling, not from the network. No distribution theory covers this design.

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# Data-driven network bandwidth: baseline and weak-network designs

| | | | $\beta$ ($\beta_0 = 1$) | | | $\alpha$ ($\alpha_0 = 0.5$) | | | |
|---|---|---|---|---|---|---|---|---|---|
| $n$ | Estimator | Fits | Bias | RMSE | Cov. | Bias | RMSE | Cov. | Merge (%) |
| *Baseline* | | | | | | | | | |
| 500 | Pairwise | 426 | 0.256 | 0.715 | 0.965 | −0.060 | 1.396 | 0.955 | – |
| | Class | 430 | 0.127 | 0.549 | 0.963 | −0.049 | 1.132 | 0.991 | 0.0 |
| | Oracle | 431 | 0.235 | 0.670 | 0.968 | −0.075 | 1.297 | 0.961 | – |
| 1000 | Pairwise | 500 | 0.152 | 0.457 | 0.950 | 0.003 | 0.880 | 0.934 | – |
| | Class | 500 | 0.084 | 0.381 | 0.974 | 0.002 | 0.757 | 0.964 | 0.0 |
| | Oracle | 500 | 0.151 | 0.459 | 0.950 | 0.000 | 0.872 | 0.940 | – |
| 2000 | Pairwise | 500 | 0.070 | 0.263 | 0.942 | 0.016 | 0.546 | 0.938 | – |
| | Class | 500 | 0.042 | 0.222 | 0.960 | −0.005 | 0.477 | 0.952 | 0.0 |
| | Oracle | 500 | 0.069 | 0.262 | 0.944 | 0.016 | 0.542 | 0.942 | – |
| *Weak network* | | | | | | | | | |
| 500 | Pairwise | 33 | −0.087 | 0.850 | 0.970 | −0.328 | 1.373 | 1.000 | – |
| | Class | 45 | 0.012 | 0.498 | 1.000 | 0.015 | 1.185 | 0.978 | 5.4 |
| | Oracle | 425 | 0.238 | 0.665 | 0.967 | −0.020 | 1.393 | 0.951 | – |
| 1000 | Pairwise | 322 | 0.278 | 0.787 | 0.941 | 0.117 | 1.657 | 0.922 | – |
| | Class | 341 | 0.100 | 0.423 | 0.947 | 0.094 | 0.881 | 0.962 | 7.8 |
| | Oracle | 500 | 0.133 | 0.451 | 0.950 | 0.018 | 0.849 | 0.948 | – |
| 2000 | Pairwise | 500 | 0.083 | 0.304 | 0.934 | −0.051 | 0.629 | 0.938 | – |
| | Class | 500 | 0.043 | 0.239 | 0.942 | −0.001 | 0.516 | 0.944 | 3.6 |
| | Oracle | 500 | 0.064 | 0.268 | 0.946 | −0.026 | 0.568 | 0.930 | – |

$R = 500$, same seeds as the main tables. Pairwise and Class use $h_1 = \hat{h}_1$, Oracle uses the population matches, and all use $h_2 = 0.5$. Fits: replications with an estimate, over which the other statistics are computed. Merge: share of replications in which a feasible class contains agents of different population classes. Doubling the constant admits false matches in the weak design (precision 0.66 at $n = 500$).

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# Glasgow: descriptive statistics

| | Wave 1 (1995) | | Wave 2 (1996) | | Wave 3 (1997) | | Range |
|---|---|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | Mean | SD | |
| Use (tobacco or cannabis) | 0.417 | 0.496 | 0.521 | 0.502 | 0.708 | 0.457 | 0–1 |
| Pocket money | 9.67 | 7.51 | 11.70 | 8.36 | 16.21 | 10.40 | 0–50 |
| Romantic relationship | 0.302 | 0.462 | 0.229 | 0.423 | 0.323 | 0.470 | 0–1 |
| Female | 0.458 | 0.501 | 0.458 | 0.501 | 0.458 | 0.501 | 0–1 |
| Age | 13.34 | 0.30 | 14.34 | 0.30 | 15.34 | 0.30 | 12.8–16.3 |
| Alcohol | 2.781 | 1.038 | 3.062 | 1.064 | 3.479 | 0.894 | 1–5 |
| Leisure | 2.616 | 0.308 | 2.748 | 0.260 | 2.736 | 0.279 | 2–4 |
| Smoking at home | 0.594 | 0.494 | 0.594 | 0.494 | 0.594 | 0.494 | 0–1 |
| Density | 0.0408 | | 0.0390 | | 0.0373 | | |
| Degree | 3.88 | 2.17 | 3.71 | 1.90 | 3.54 | 1.80 | 0–10 |

- 96 students, 3 waves, 288 observations. Pocket money is monthly, in pounds. Leisure is the mean of fifteen items coded from 1 (most days) to 4 (less often or never), so a higher value means less activity.
- The outcome equals one for any tobacco use or any cannabis use (including "tried once"), so the estimand is persistence in the recorded composite status; tobacco-only is a robustness check.
- Pocket money and leisure take 47 and 27 distinct values, so they are effectively continuous. Female and smoking at home difference out of the conditional fits. Age rises by one year for every student, so it cancels in the pairwise and class fits, and HK omits it.

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# Glasgow: descriptive fits, any-friend network

| | (1) Naive | (2) Controls | (3) HK | (4) Pairwise | (5) Class |
|---|---|---|---|---|---|
| Intercept | −4.931 | −8.039 | — | — | — |
| | (2.779) | (3.523) | | | |
| Pocket money | 0.014 | 0.010 | 0.062 | 0.099 | 0.059 |
| | (0.019) | (0.020) | (0.046) | (0.050) | (0.064) |
| Romantic relationship | 0.815 | 0.543 | 2.423 | 4.943 | 2.741 |
| | (0.346) | (0.381) | (1.731) | (2.272) | (2.365) |
| Alcohol | 0.808 | 0.800 | 1.501 | 3.253 | 1.698 |
| | (0.174) | (0.185) | (0.414) | (0.858) | (0.988) |
| Leisure | 0.371 | 0.087 | 2.436 | 5.403 | 2.881 |
| | (0.532) | (0.564) | (1.712) | (2.740) | (2.713) |
| Female | −0.773 | −0.021 | — | — | — |
| | (0.315) | (0.638) | | | |
| Age | 0.070 | 0.285 | — | — | — |
| | (0.195) | (0.245) | | | |
| Smoking at home | 0.138 | 0.207 | — | — | — |
| | (0.303) | (0.324) | | | |
| Lagged use ($\hat{\alpha}$) | 2.222 | 2.101 | 1.827 | 1.413 | 0.893 |
| | (0.420) | (0.444) | (0.668) | (1.368) | (1.536) |
| Peer average covariates | — | yes | — | — | — |
| Firth penalization | — | — | yes | yes | yes |
| Observations | 288 | 288 | 22 agents | 96 pairs | 22 agents |

Standard errors for scale only; no significance levels. Columns (3) and (4): sandwich clustered on the 22 switchers; column (5): inverse class-conditional information. No unpenalized conditional likelihood is separated; the unpenalized estimates of $\alpha$ are 3.884 (HK), 2.281 (pairwise), and 2.134 (class), and penalization shrinks every conditional coefficient by a third to a half, or more for the class fit, keeping the sign pattern. Columns (4) and (5) do not use the network.

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# Glasgow: robustness to the outcome and the initial condition

| | Baseline (any use, $y_{i0} = 0$) | Tobacco only | $y_{i0} = 1$ for wave-1 regular smokers | $y_{i0} = y_{i1}$ |
|---|---|---|---|---|
| Students / switchers / pairs | 96 / 22 / 96 | 97 / 16 / 48 | 96 / 22 / 96 | 96 / 22 / 0 |
| $\hat{\alpha}$, Naive | 2.222 | 2.816 | 2.438 | 3.893 |
| $\hat{\alpha}$, HK | $1.827$ $(0.668)$ | $1.001^{\ddagger}$ $(0.874)$ | $1.827$ $(0.668)$ | $2.873^{\dagger}$ $(0.502)$ |
| $\hat{\alpha}$, Pairwise | $1.413$ $(1.368)$ | $-1.983^{\dagger}$ $(1.683)$ | $1.413$ $(1.368)$ | — |
| $\hat{\alpha}$, Class | $0.893$ $(1.536)$ | $-1.274^{\dagger}$ $(2.981)$ | $0.893$ $(1.536)$ | — |

$^{\dagger}$ unpenalized likelihood completely separated (certified by a linear program): the penalized value carries no information. $^{\ddagger}$ quasi-completely separated: only one of the 16 switchers changes romantic status.

- **Tobacco only:** 16 switchers and 48 pairs; HK's estimate drops to 1.00, and the pairwise and class likelihoods are separated, so positive state dependence in the conditional fits rests on the composite outcome.
- **Regular smokers used before wave 1:** the conditional fits do not move, since none of those ten students is among the 22 switchers.
- **Every wave-1 user used a year earlier:** no pairs remain, because lag matching needs $y_{i0} = y_{j0}$ while discordance needs $y_{i1} \neq y_{j1}$. The initial condition is load-bearing, and a test of the mechanism needs at least four waves.

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# Glasgow: network-localized estimates

| | Any friend | | | Best friend |
|---|---|---|---|---|
| | $q = 0.10$ | $q = 0.20$ | $q = 0.30$ | $q = 0.10$ |
| *Separation diagnostics (unpenalized likelihood)* | | | | |
| Share of fitted probabilities outside $[0.01, 0.99]$ | 1.000 | 1.000 | 1.000 | 1.000 |
| Max \|fitted index\| at optimizer stop | 38.2 | 67.2 | 51.7 | 53.9 |
| Max. min. signed margin ($\|b\| = 1$) | 1.540 | 0.566 | 0.483 | 0.729 |
| MLE exists | no | no | no | no |
| *Firth-penalized estimates* | | | | |
| Pocket money | $-0.005$ $(0.047)$ | $-0.009$ $(0.040)$ | $-0.004$ $(0.038)$ | $0.009$ $(0.193)$ |
| Romantic relationship | $-0.658$ $(2.481)$ | $-0.388$ $(1.765)$ | $-0.786$ $(1.628)$ | $0.126$ $(1.061)$ |
| Alcohol | $0.317$ $(0.513)$ | $0.625$ $(0.413)$ | $0.824$ $(0.397)$ | $0.847$ $(0.600)$ |
| Leisure | $0.867$ $(2.797)$ | $0.820$ $(1.975)$ | $1.502$ $(1.820)$ | $0.179$ $(3.188)$ |
| Lagged use ($\hat{\alpha}$) | $0.176$ $(1.514)$ | $-0.217$ $(0.996)$ | $-0.403$ $(0.980)$ | $0.876$ $(1.184)$ |
| Pairs with positive weight | 9 | 21 | 26 | 14 |
| Pair-weight effective sample size | 7.0 | 17.1 | 23.0 | 14.0 |

$K_2$ omitted throughout. A positive margin certifies complete separation; values in parentheses are the unpenalized sandwich at the penalized estimates, for scale only. The $q = 0.05$ bandwidth is infeasible (5 pairs). Best-friend average degree is 0.81, 0.67, and 0.81 across waves. At each bandwidth the class estimator has a single class with both orderings, with 9 to 15 of the 22 switchers, and its likelihood is separated too (margins 0.20 to 0.34).

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# Network measurement in the Glasgow data

The model takes links to be undirected, formed from conditionally independent dyadic shocks, observed without error, and unconstrained in number; the asymptotics are those of a dense graphon. The Glasgow nominations depart from this in three ways.

- **Nomination cap and the OR rule.** Students name at most six friends, so one nomination constrains another and the links of a student are dependent even given the characteristics, not only truncated for high-degree students. Reciprocated-only links would not remove the dependence and would make the network sparser. The OR rule turns an unreciprocated nomination into a link, whether or not it reflects a friendship.
- **Bounded degrees.** Average degree is below four in the any-friend network and below one in the best-friend network, far from the dense regime in which codegrees are informative; a larger school with bounded degrees would not help by itself.
- **Induced subgraphs.** The networks are those of the 96 complete cases; nominations to and from the other 64 students are discarded, with no correction.
- None of this enters the unlocalized conditional fits, which do not use the network; all of it enters the localized estimates.

These features motivate the next project: network-type matching with directed, censored, and misreported networks.

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# Appendix C: ordered outcomes

Ordered outcome $Y_{it} \in \{1, \ldots, J\}$ with state dependence through a known cutoff $c$ (Muris, Raposo, and Vandoros, 2025; Honoré, Muris, and Weidner, 2025):

$$Y_{it}^* = X_{it}'\beta + \alpha^{\text{ord}}\, \mathbb{1}\{Y_{i,t-1} \geq c\} + \lambda(w_{it}) + \mu_i - \varepsilon_{it}, \qquad Y_{it} = j \;\text{ if }\; \varsigma_j \leq Y_{it}^* < \varsigma_{j+1}.$$

Dichotomizing at the cutoff, with $\varsigma_c = 0$, gives a binary dynamic logit of exactly the form studied here:

$$B_{it}(c) = \mathbb{1}\{Y_{it} \geq c\} = \mathbb{1}\big\{X_{it}'\beta + \alpha^{\text{ord}} B_{i,t-1}(c) + \lambda(w_{it}) + \mu_i - \varepsilon_{it} \geq 0\big\}.$$

- The estimator replaces $y_{it}$ by $B_{it}(c)$ throughout, so all results apply to the dichotomized model under the same assumptions, provided the link function depends on the lagged outcome only through $B_{i,t-1}(c)$.
- It uses only the cutoff $c$; a composite version pooling all thresholds, whose dichotomized errors share the same $\varepsilon_{it}$, is left for future work.
- In the Glasgow data, tobacco use is classified as non-smoker, occasional, or regular; with $c = 2$ the reduction separates non-use from any use. This tobacco-only illustration is distinct from the composite outcome of the application.

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